Human-Computer Interaction

8. Data Analysis

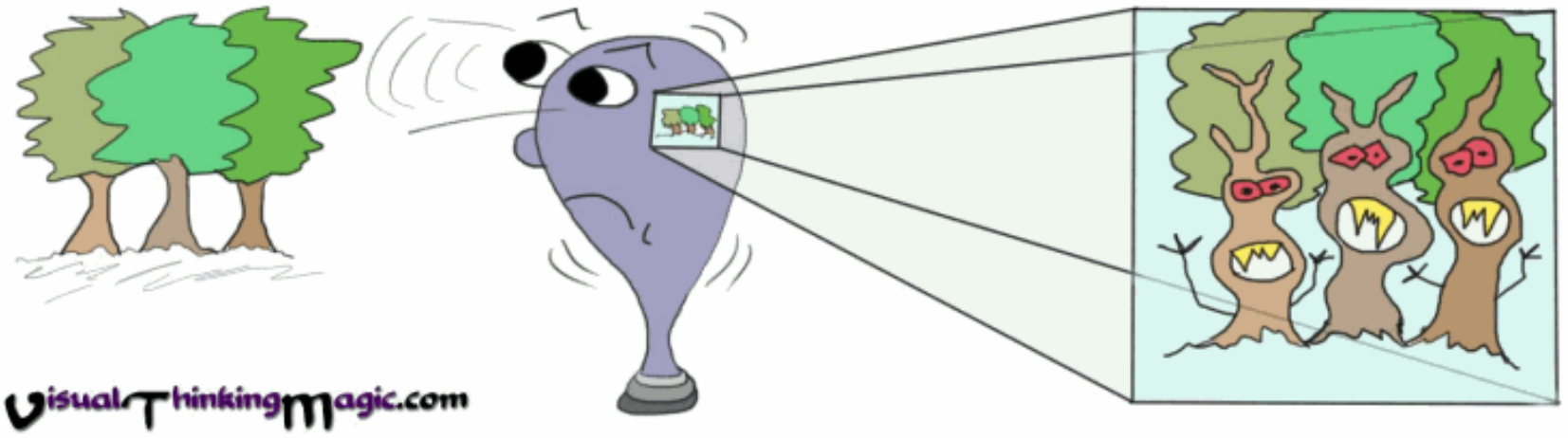


Last week

- Mental Model
- Model human processor
- Perceptual processor
- Working memory
- Cognitive processor
- Motor processor

Recap: Mental model

"How your mind thinks that something works"

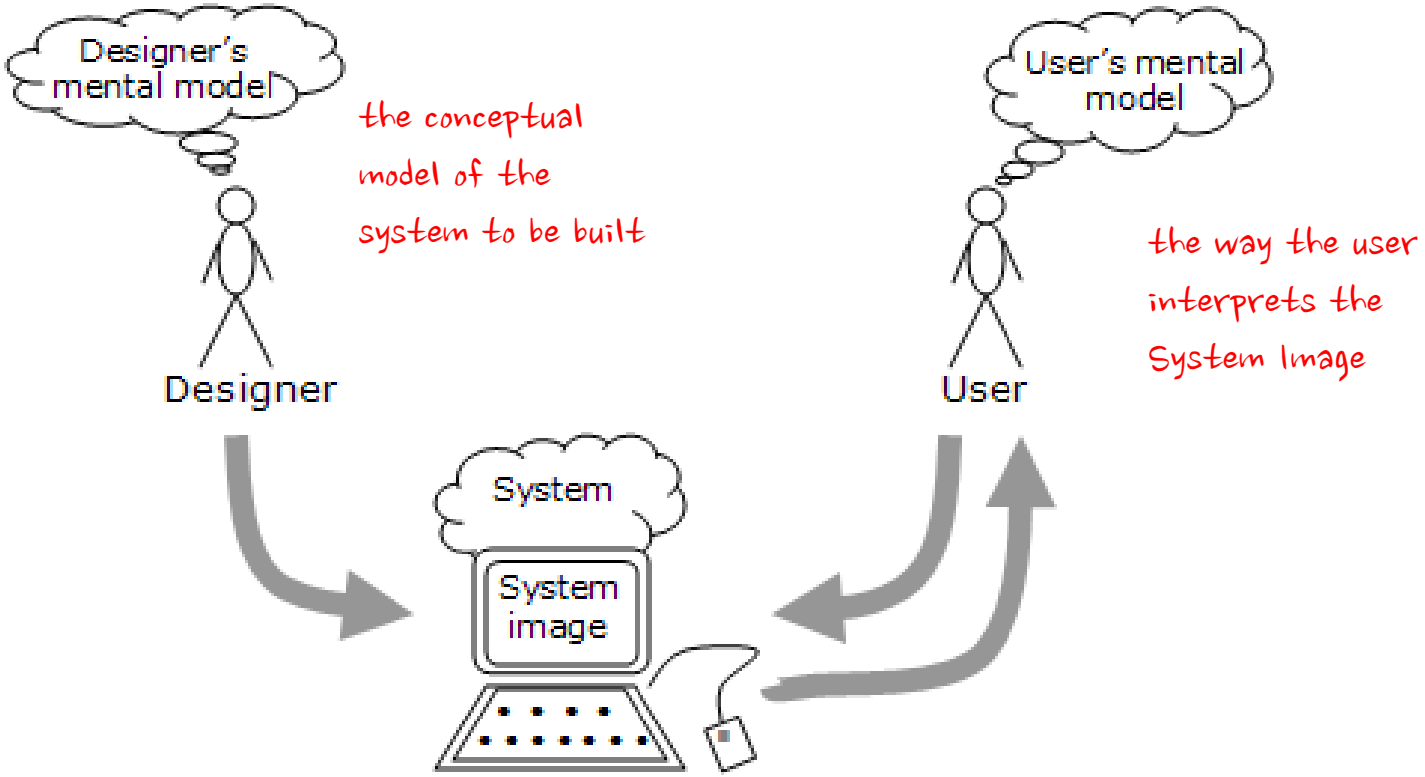


Recap: Mental model

"The models people have of themselves, others, the environment and the things with which they interact. People form mental models through experience, training and instruction"

- An explanation of someone's thought process about how something works in the real world
- Enable people to reason about a system
- Affect the way we see and interpret reality
- When users approach an unfamiliar system, they subconsciously refer to their mental model, User's model

Recap: User's vs. designer's model



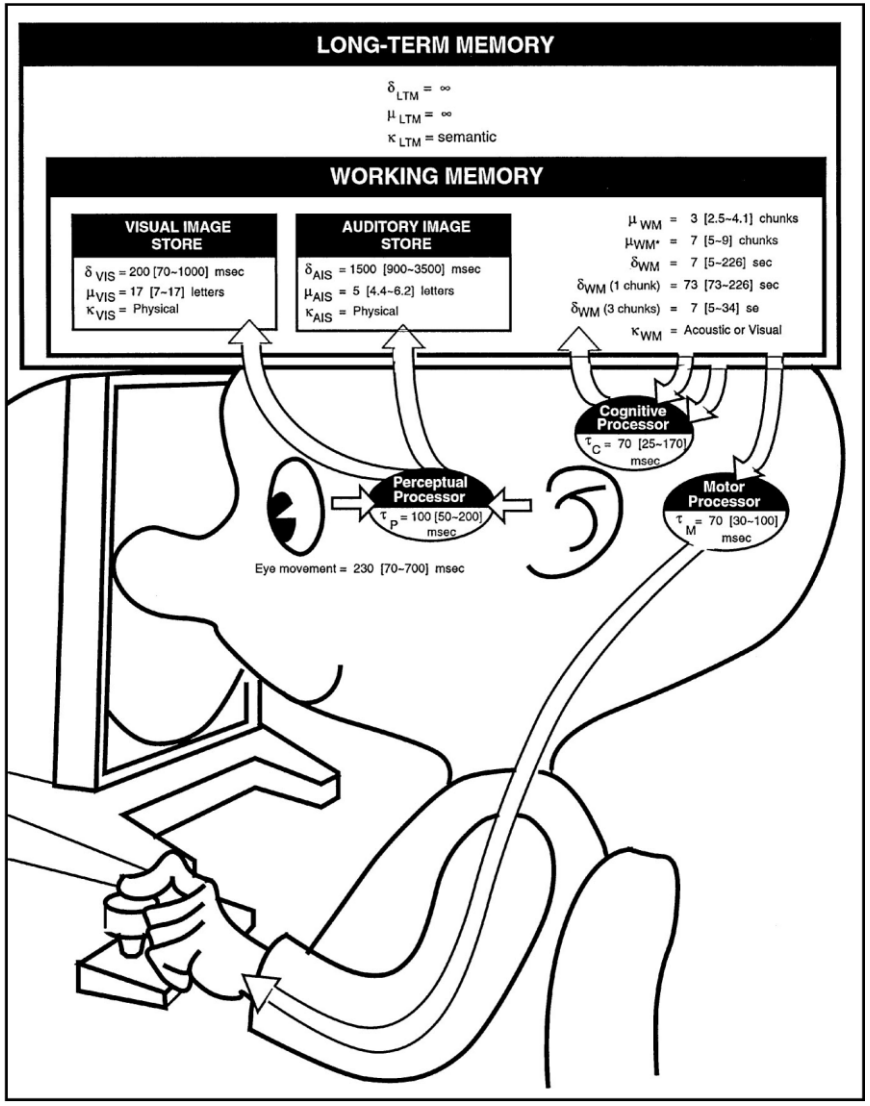
As a designer, you want to use the system's UI to accurately convey your mental model of the system to the user.

Recap: The Model Human Processor

- Calculate how long it takes to perform a certain task
- Consider humans as information processing systems

- Consist of three interacting processors
  1. Perceptual processor: Responsible for transforming external environment into a form that cognitive system can process
  2. Cognitive processor: Uses contents of working memory and long-term memory to make decisions and schedule actions with motor system
  3. Motor processor: Translates thoughts into actions

Recap: The Model Human Processor



Recap: Visual Processing Properties

Pre-attentive

- A limited set of visual properties that are detected very rapidly and accurately by the low-level visual system without our conscious action.
- It allows the designer to grab the attention of the user without their input or effort.

Change blindness

- A perceptual phenomenon that occurs when a change in a visual stimulus is introduced and the observer does not notice it

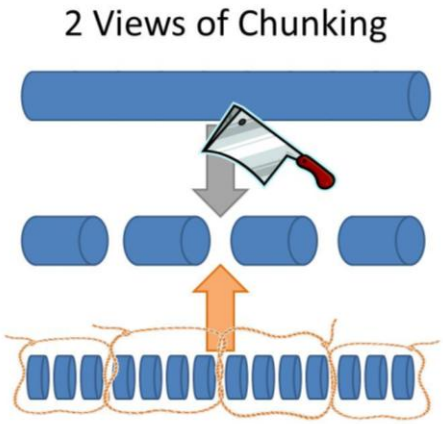
Selective attention (Tunnel Vision)

- The act of focusing on a particular object for a period of time while simultaneously ignoring irrelevant information that is also occurring

Recap: The Magic number 7 ± 2

- George Miller's theory of how much information people can remember
- People's immediate memory capacity is very limited
- The number of objects an average human can hold in working memory (short-term) is 7 ± 2

Recap: Chunking



- Breaking up long strings of information into units
- The primary purpose of chunking is the enhancement of working memory

When to use chunking:

- Ideal when specific information must be memorized for later use
- Effective use of chunking can improve the usability and effectiveness of an information system

When not to use chunking:

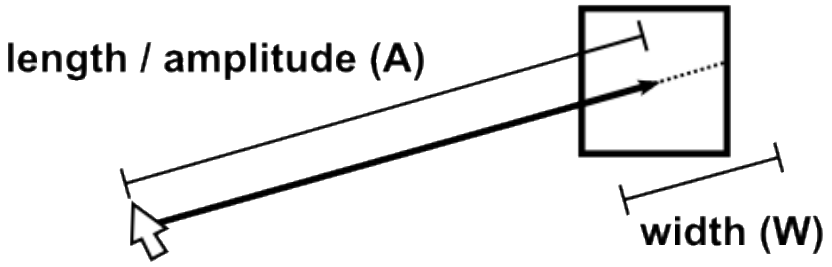
- Should not be used when the information must be searched, scanned, or analyzed
- Should not be used as an argument for improved simplicity, legibility, or uncluttered page design

Recap: Fitts's law

- Fitts's law predict the amount of time taken to move to and select a target
- It is faster to hit larger targets close to you than smaller targets further from you
- T (Travel time to hit the target) increases as the distance to the target (A) increases
- T decreases as the size of the target (W) increases

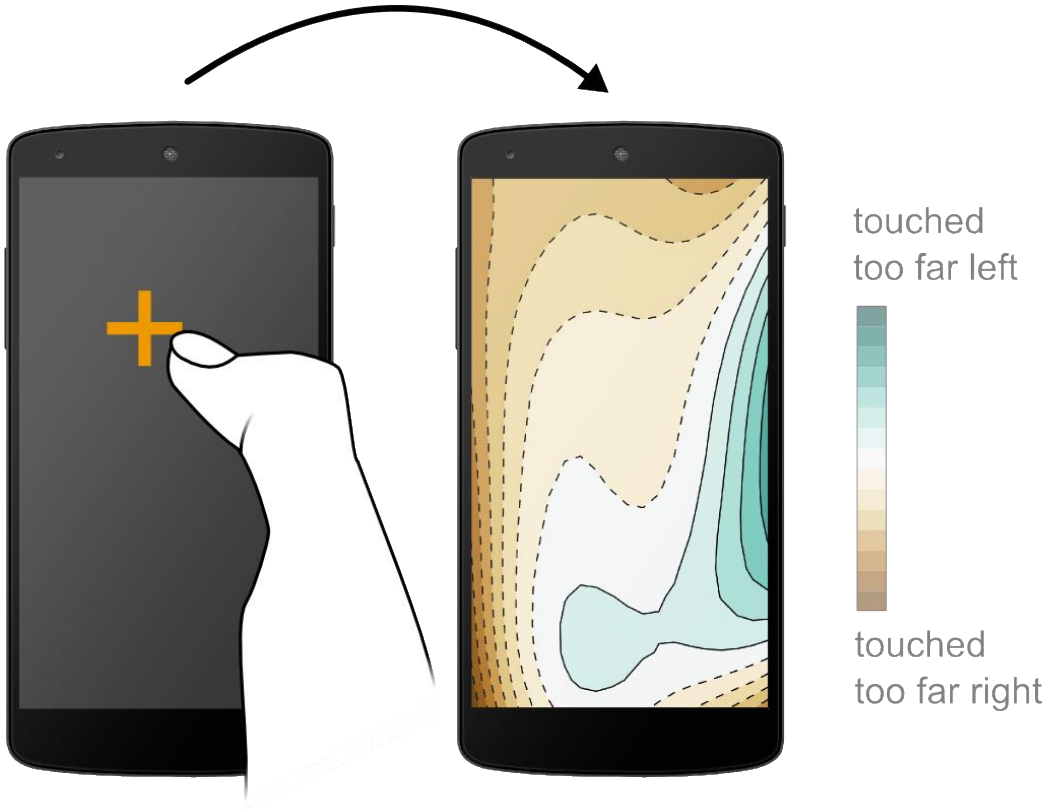
Ways to apply Fitts's law in UI design:

- Bring items closer to the cursor
- Increase the target size
- Exploit the edges



Recap: Fitts's law in Mobile UI

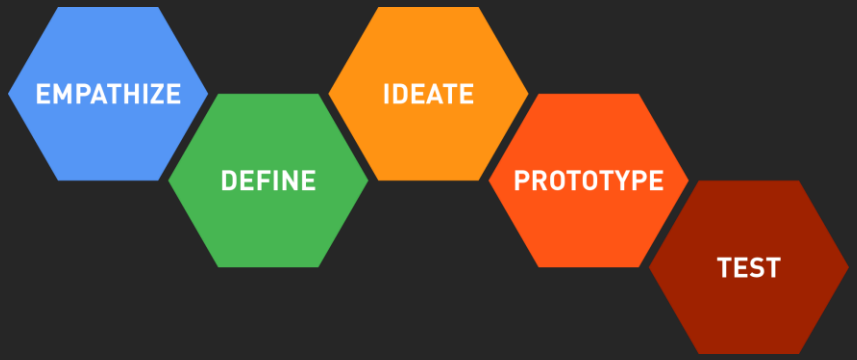
- No benefit on device edges
- How device is held is important



Today's agenda

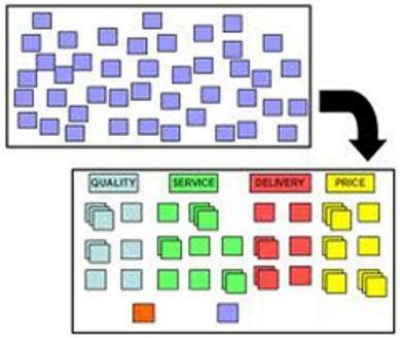
- Data Analysis
  - Affinity Diagram
  - Work Models
  - Survey Report





Affinity diagram

Affinity diagram



"a UX designer's tool, used to capture and synthesize qualitative data"

The affinity diagram brings together issues and insights across all users into a hierarchical diagram to reveal the scope of the problem.

- Gathers large amounts of language data (ideas, opinions, issues) and organizes them into groupings based on their natural relationships
- A single picture of the customer population a design will address
- Shows how the work hangs together rather than breaking it up in lists and what matters
- Used to brainstorm and generate ideas

https://zhuanlan.zhihu.com/p/496643551

Affinity Diagram

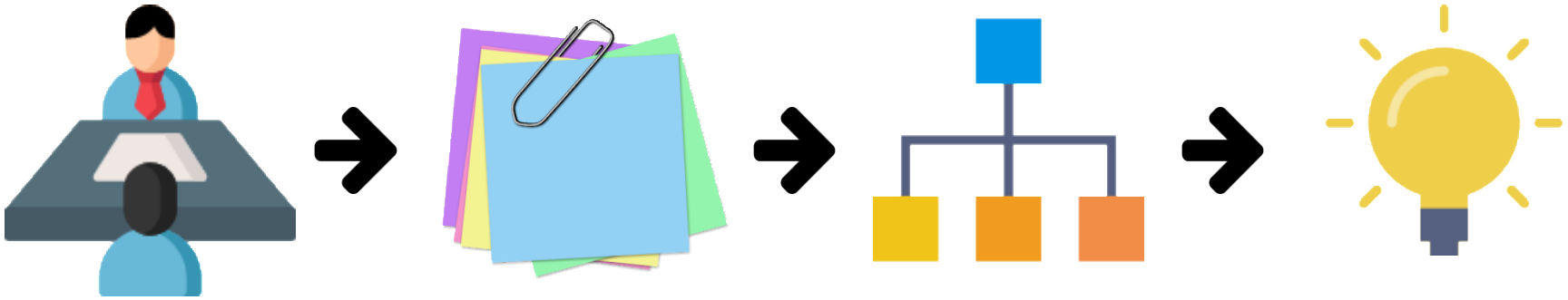
- Affinity Diagrams are generated during group session
- Each observation/idea/note is copied to a post-it
- Notes are hierarchically organized into themes, based on the focus of the project
- Usually done in a chaotic fashion, with design team members running back and forth with post-its and yelling ideas to each other

Reading responses

- They don't categorize the stickies, they let the stickies categorize themselves. This is important because then we can actually get insights instead of place data into already set assumptions... I thought this was very helpful because it allowed me to find more categories than I would've found if I grouped them originally in the categories I thought we had. - Anchalee

- It's important to work as a team - who is also structured - to receive different perceptions and viewpoints of the users' data because after all, you could never design for just one person nor could you, I think, design something as a one man team without input and ideas from other designers. - Mary

Steps: Creating an Affinity Diagram



Collect Data

Extract raw data

Synthesize data

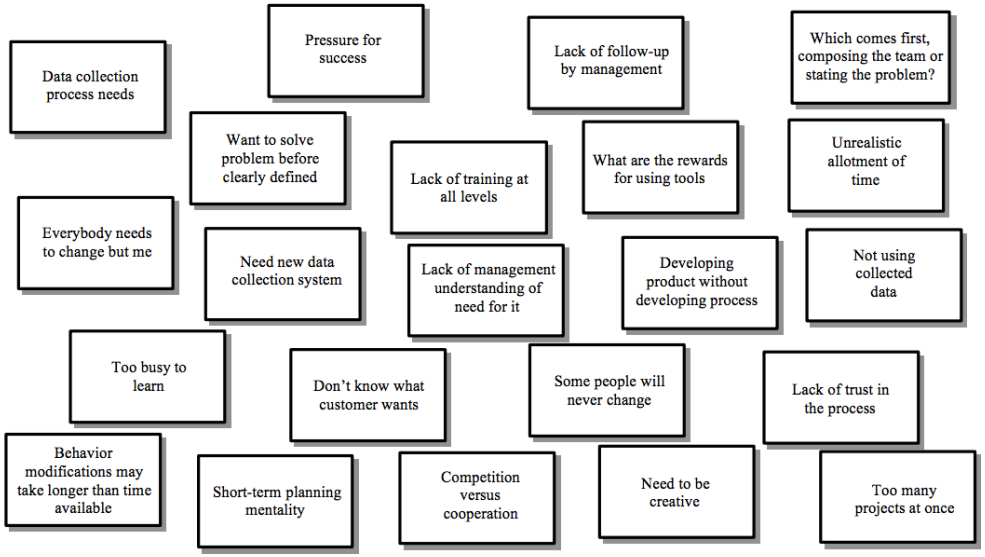
Understand

1.Collect Data

- Interviews
- Surveys
- Observations
- Field notes
- …

2. Extract raw data

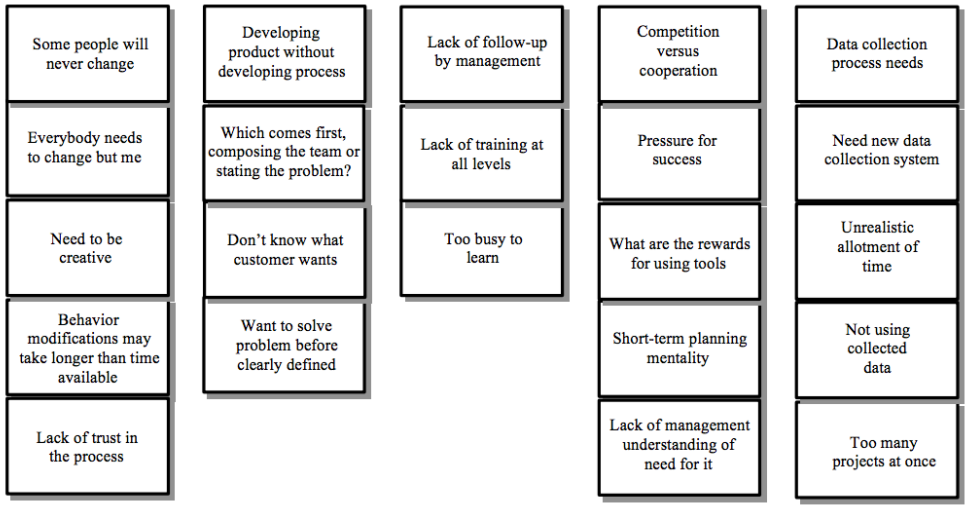
- Record all of your interviews and re-listen to them as you transfer the insights on to post-it notes
- Start each post-it note from the perspective of the user
- For each person that you spoke with, choose a different color post-it note



3. Synthesis

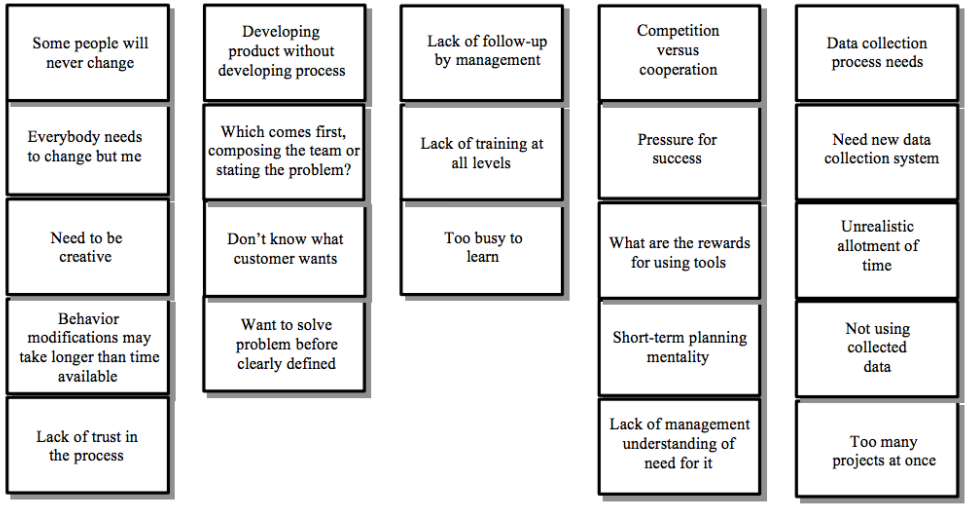
Arrange them into clusters that make sense to you! In another word, sort the sticky notes into groups, such as (but not limited to):

- Biographical information
- Frustrations
- Interactions/touch points
- goals/motivations



3. Synthesis

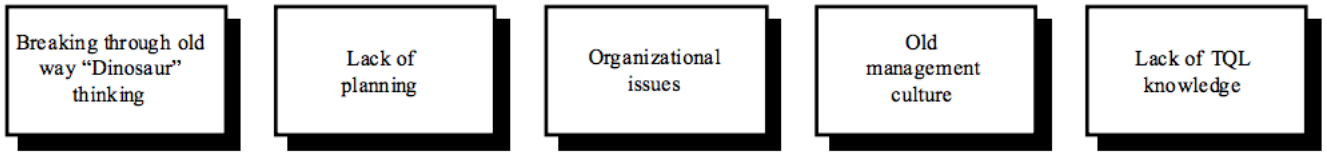
- Start by looking for two ideas that seem related in some way. Place them together in a column off to one side
- Look for ideas that are related to those you've already set aside and add them to that group
- Look for other ideas that are related to each other and establish new groups



4. Create header cards

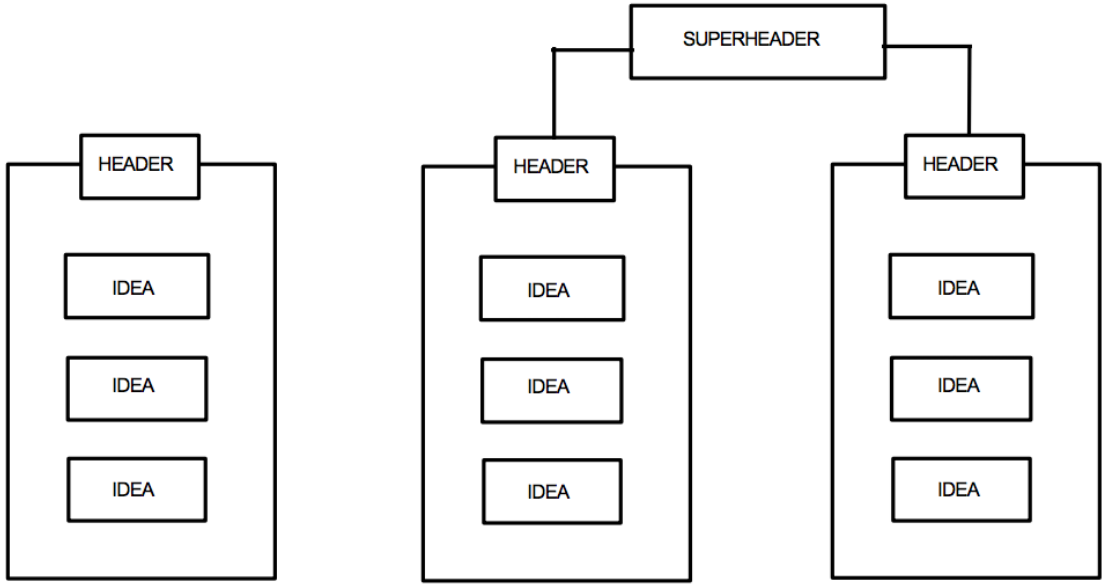
Header: An idea that captures the essential link among the ideas contained in a group of cards

- Finding already existing cards within the groups that will serve well as headers and placing them at the top of the group of related cards
- Alternatively, discussing and agreeing on the wording of cards created specifically to be headers
- Discovering a relationship among two or more groups and arranging them in columns under a super-header. The same rules apply for super-headers as for regular header cards
- It should be concise and have a clear meaning!



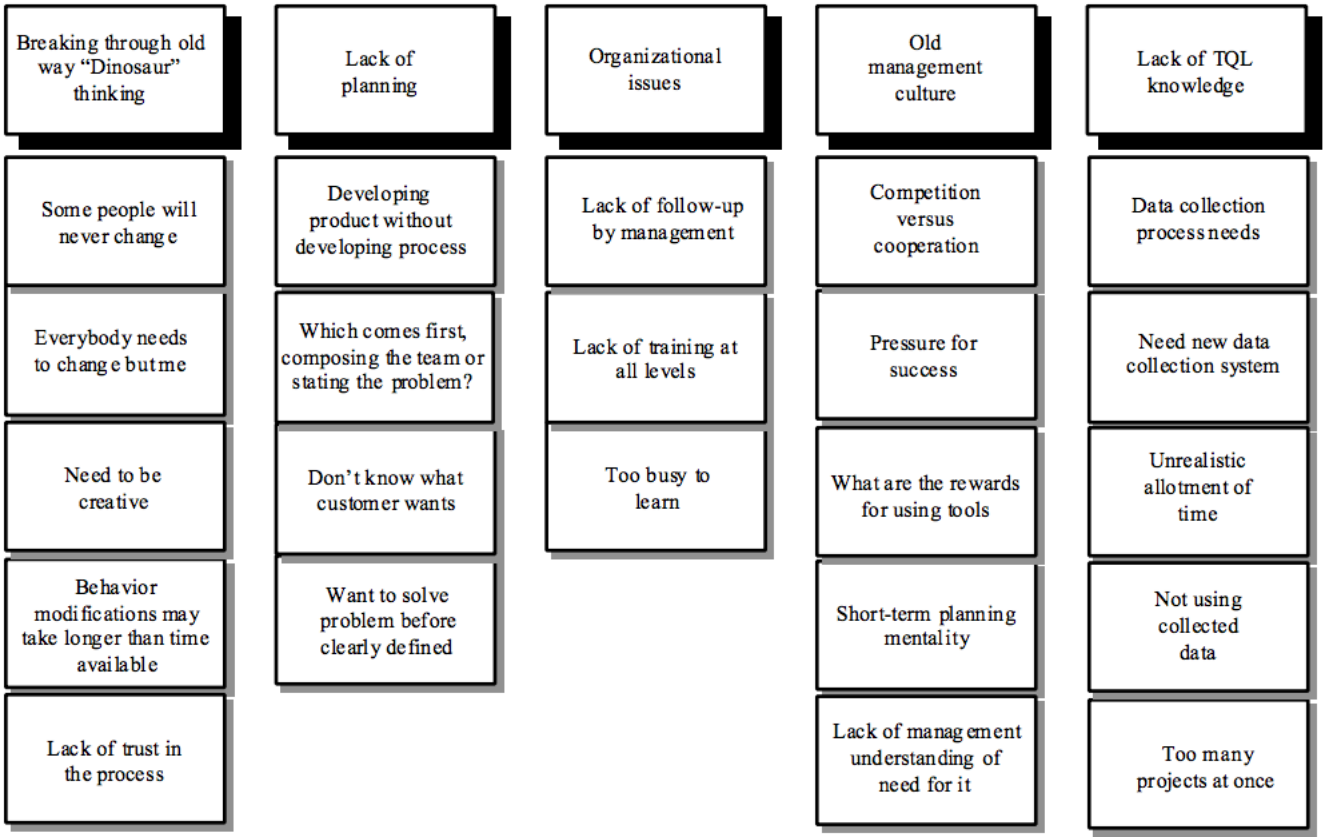
5. Draw finished diagram

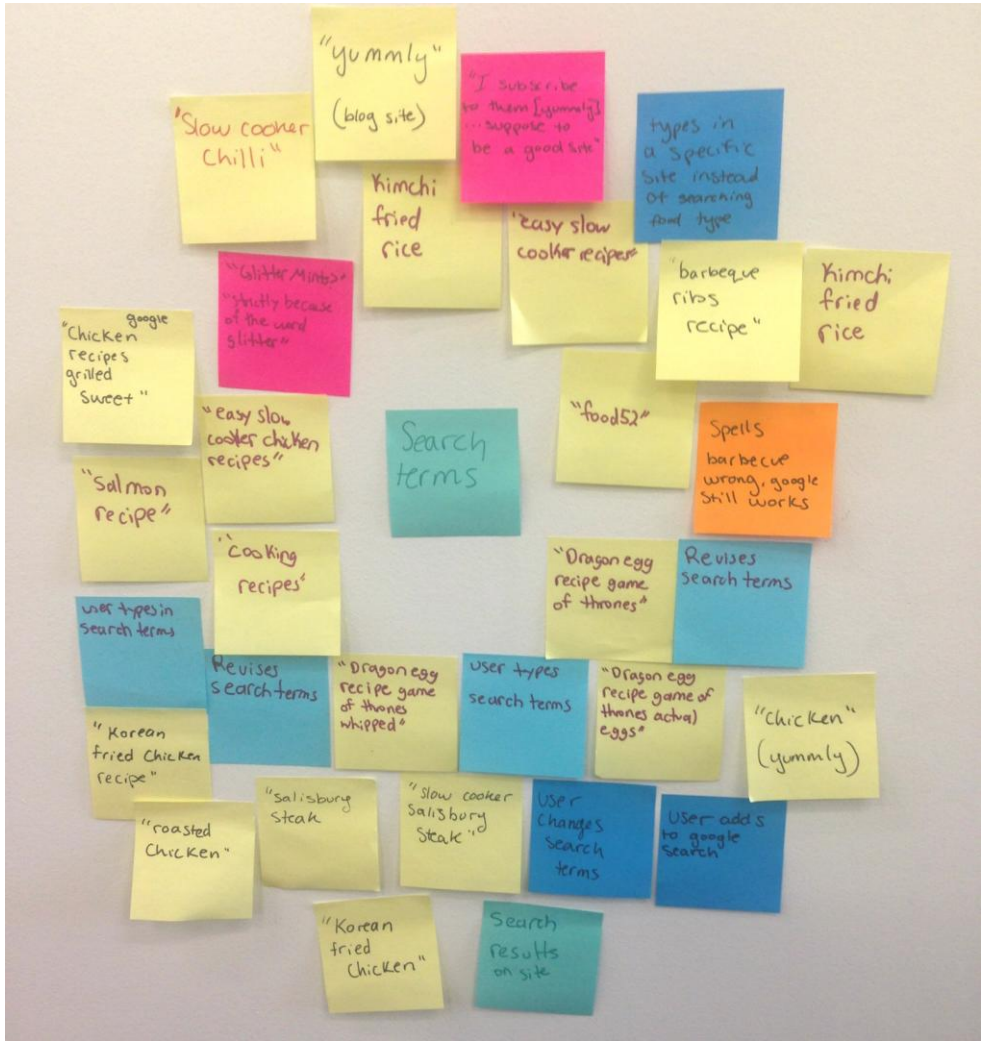
- Write a problem statement at the top of the diagram
- Place header and super-header cards above the groups of ideas
- Review and clarify the ideas and groupings
- Document the finished Affinity Diagram

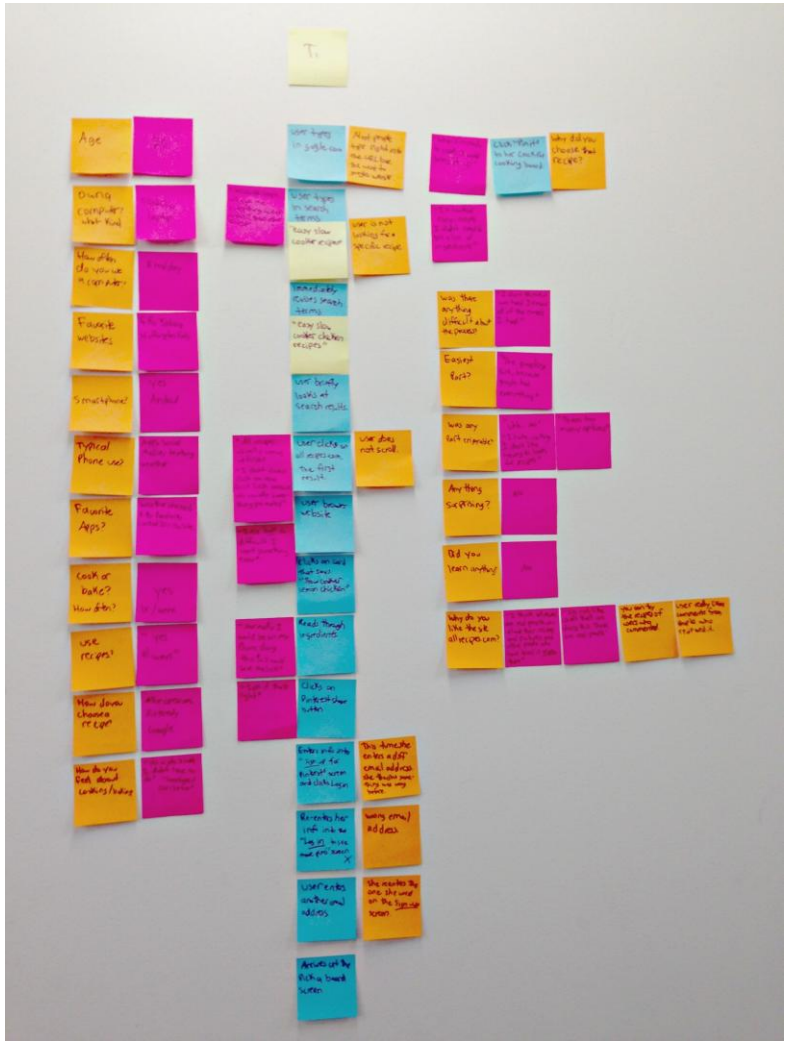


5. Draw finished diagram

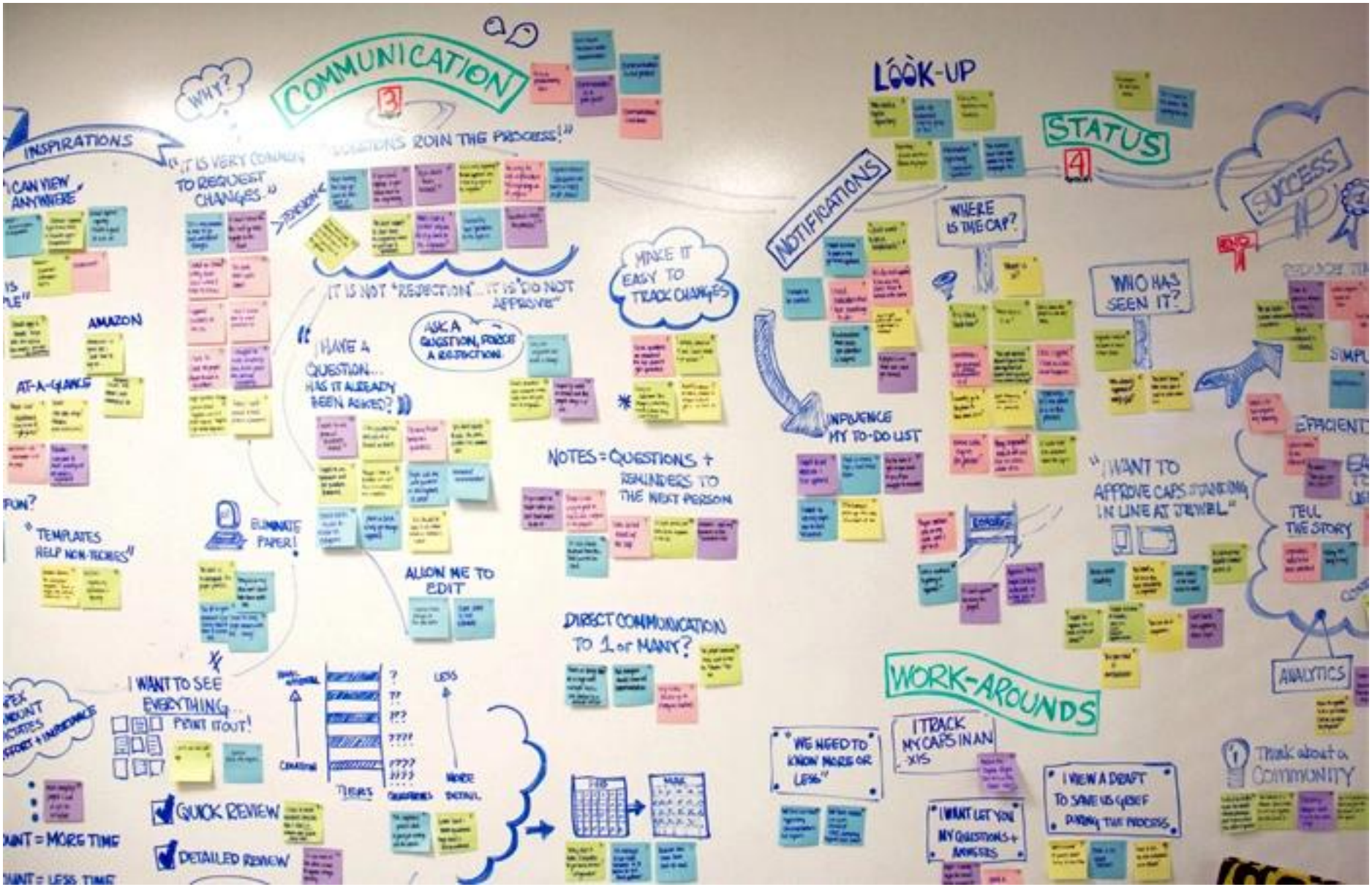
Issues in implementing continuous process improvement



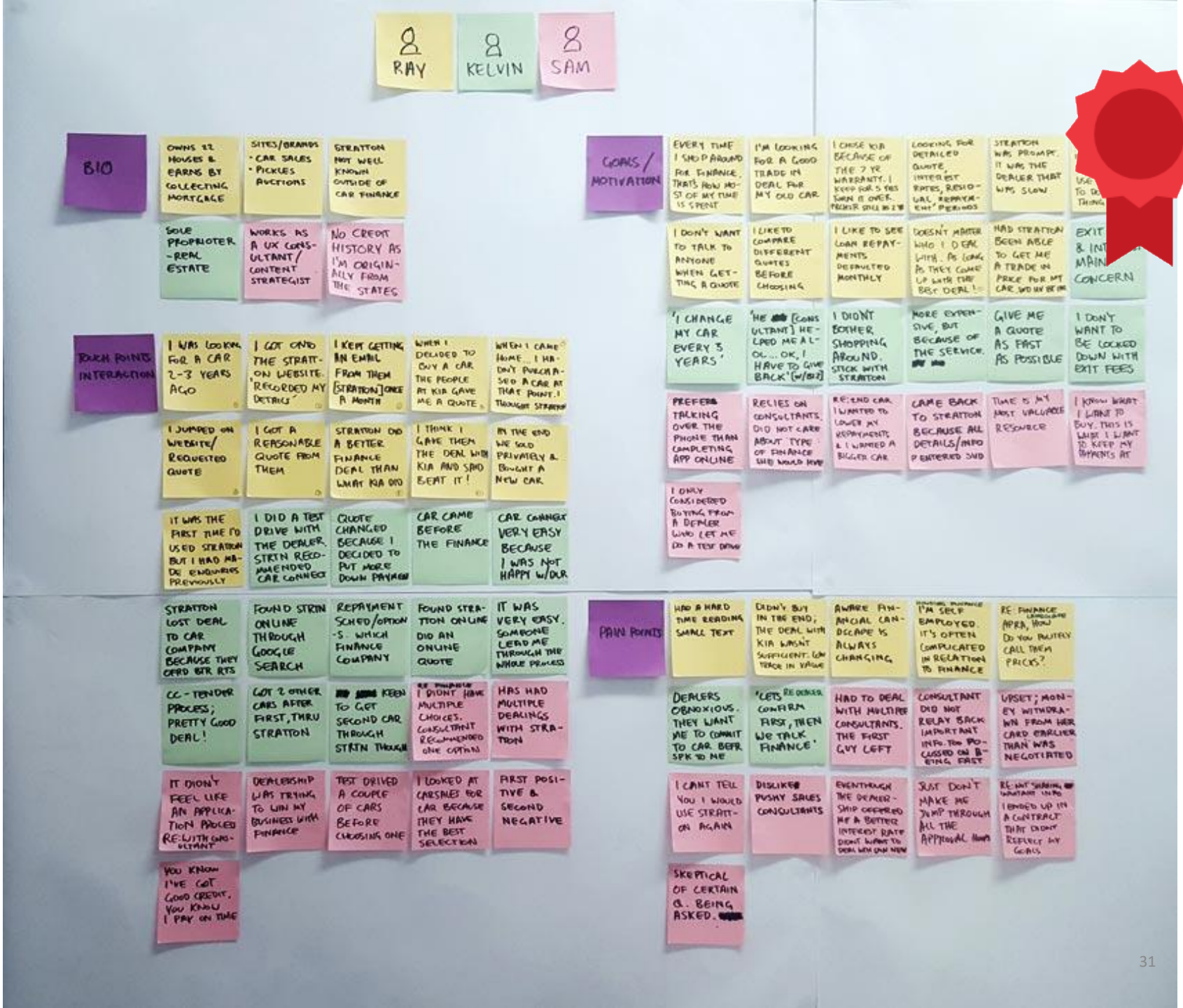

"yummly" (blog site)
"I subscribe to them [yummly] ... suppose to be a good site"
"Slow cooker Chilli"
types in a specific site instead of searching food type
kimchi fried rice
"easy slow cooker recipes"
"Glitter Mints?" "strictly because of the word glitter"
"barbeque ribs recipe"
kimchi fried rice
google "Chicken recipes grilled sweet"
"easy slow cooker chicken recipes"
Search terms
"food52"
Spells barbecue wrong, google still works
"Salmon recipe"
"cooking recipes"
"Dragon egg recipe game of thrones"
Revises search terms
user types in search terms
Revises search terms
"Dragon egg recipe game of thrones whipped"
User types search terms
"Dragon egg recipe game of thrones actual eggs"
"Chicken" (yummly)
"Korean fried Chicken recipe"
"roasted Chicken"
"salisbury steak
"slow cooker salisbury steak"
User Changes search terms
User adds to google search
"Korean fried Chicken"
Search results on site

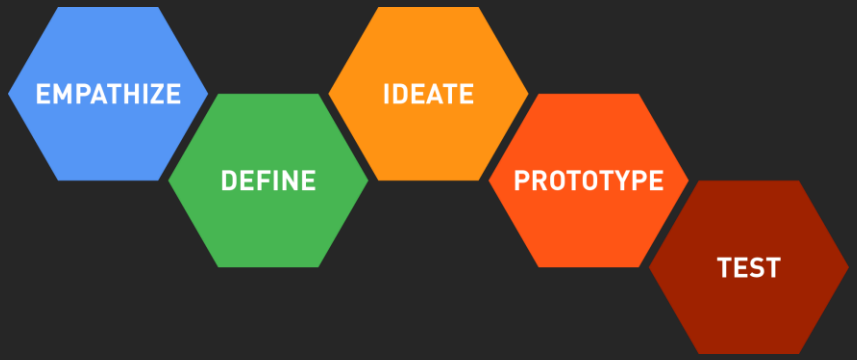

Age
Typical Phone use
Favorite Apps?
Favorite websites
Smartphone?
Cook or bake?
How often do you use recipes?
Owning computer? what kind
How do you find dessert cookies/bakes
User types in search terms
"easy slow cooker recipes"
"easy slow cooker chicken recipes"
User types in google
User types website
User clicks on Pinterest share button
User clicks on first result
User does not scroll
Click through ingredients
Was there anything difficult about the process?
Easiest Part?
Was any part stressful?
Anything surprising?
Did you learn anything?
Why do you like the UK all recipes site?
Why did you choose this RECIPE?
Click through to the cooking board
User enters email address
User enters another email address
Articles set by Pick a board screen



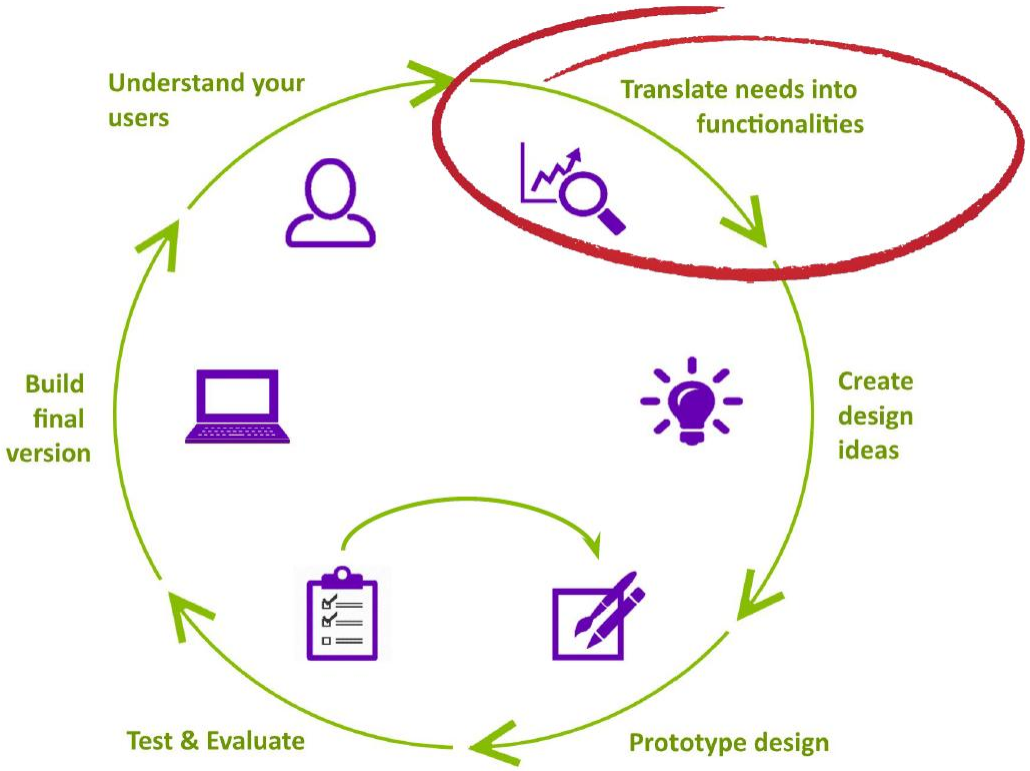

COMMUNICATION
3
WHY?
INSPIRATIONS
"I CAN VIEW ANYWHERE"
AMAZON
AT-A-GLANCE
"TEMPLATES HELP NON-TECHIES"
"IT IS VERY COMMON TO REQUEST CHANGES"
"QUESTIONS JOIN THE PROCESS!"
"IT IS NOT "REJECTION"... IT IS "DO NOT APPROVE""
"I HAVE A QUESTION... HAS IT ALREADY BEEN ASKED?"
ASK A QUESTION, FORCE A REJECTION
MAKE IT EASY TO TRACK CHANGES
ELIMINATE PAPER!
ALLOW ME TO EDIT
NOTES = QUESTIONS + REMINDERS TO THE NEXT PERSON
DIRECT COMMUNICATION TO 1 or MANY?
"I WANT TO SEE EVERYTHING... PRINT IT OUT!"
QUICK REVIEW
DETAILED REVIEW
LESS
MORE DETAIL
CREATIVE
TIERS
"MORE TIME"
LOOK-UP
STATUS
4
NOTIFICATIONS
WHERE IS THE CAP?
WHO HAS SEEN IT?
INFLUENCE MY TO-DO LIST
"I WANT TO APPROVE CAPS STANDING IN LINE AT JEWEL"
SUCCESS
EFFICIENT
TELL THE STORY
ANALYTICS
WORK-AROUNDS
"WE NEED TO KNOW MORE OR LESS"
I TRACK MY CAPS IN AN .XLS
I WANT LET YOU MY QUESTIONS + ANSWERS
I VIEW A DRAFT TO SAVE US TIME PRINTING THE PROCESS
Think about a Community


I DIDN'T BOTHER SHOPPING AROUND. STICK WITH STRATTON
GIVE ME A QUOTE AS FAST AS POSSIBLE
DEALERS OBNOXIOUS. THEY WANT ME TO COMMIT TO CAR BEFR SPK TO ME
'LETS CONFIRM FIRST, THEN WE TALK FINANCE'
CAR CONNECT VERY EASY BECAUSE I WAS NOT HAPPY W/DLR
I DON'T WANT TO BE LOCKED DOWN WITH EXIT FEES
REPAYMENT SCHED/OPTION -S. WHICH FINANCE COMPANY
QUOTE CHANGED BECAUSE I DECIDED TO PUT MORE DOWN PAYMT
SOLE PROPRIOTER -REAL ESTATE
FOUND STRTN ONLINE THROUGH GOOGLE SEARCH
I DID A TEST DRIVE WITH THE DEALER. STRTN RECOMMENDED CAR CONNECT
'I CHANGE MY CAR EVERY 3 YEARS'
'HE [CONSULTANT] HELPED ME ALOT... OK, I HAVE TO GIVE BACK' [W/BIZ]
HE WAS KEEN TO GET SECOND CAR THROUGH STRTN THOUGH
STRATTON LOST DEAL TO CAR COMPANY BECAUSE THEY OFFD BTR RTS
EXIT FEES & INTEREST MAIN CONCERN
I GOT A REASONABLE QUOTE FROM THEM
DOESN'T MATTER WHO I DEAL WITH - AS LONG AS THEY COME UP WITH THE BEST DEAL!
CC - TENDER PROCESS; PRETTY GOOD DEAL!
MORE EXPENSIVE, BUT BECAUSE OF THE SERVICE.
CAR CAME BEFORE THE FINANCE
GOT 2 OTHER CARS AFTER FIRST, THRU STRATTON
FOUND STRATTON ONLINE DID AN ONLINE QUOTE
IT WAS VERY EASY. SOMEONE LEAD ME THROUGH THE WHOLE PROCESS
LOOKING FOR DETAILED QUOTE, INTEREST RATES, RESIDUAL, REPAYMENT PERIODS
I JUMPED ON WEBSITE/ REQUESTED QUOTE
I GOT ONTO THE STRATTON WEBSITE. RECORDED MY DETAILS
RE: FINANCE LANGUAGE APRA, HOW DO YOU POLITELY CALL THEM PRICKS?
I LIKE TO SEE LOAN REPAYMENTS DEFAULTED MONTHLY
OWNS 22 HOUSES & EARNS BY COLLECTING MORTGAGE
I CHOSE KIA BECAUSE OF THE 7 YR WARRANTY. I KEEP FOR 5 YRS TURN IT OVER. PRICHSR STILL IN 2'ND
SITES/BRANDS - CAR SALES - PICKLES AUCTIONS
HAD A HARD TIME READING SMALL TEXT
I'M LOOKING FOR A GOOD TRADE IN DEAL FOR MY OLD CAR
I THINK I GAVE THEM THE DEAL WITH KIA AND SAID BEAT IT!
I WAS LOOKING FOR A CAR 2-3 YEARS AGO
AWARE FINANCIAL CANDSCAPE IS ALWAYS CHANGING
I LIKE TO COMPARE DIFFERENT QUOTES BEFORE CHOOSING
I'M SELF EMPLOYED. IT'S OFTEN COMPLICATED IN RELATION TO FINANCE
I DON'T WANT TO TALK TO ANYONE WHEN GETTING A QUOTE
EVERY TIME I SHOP AROUND FOR FINANCE THAT'S HOW MOST OF MY TIME IS SPENT
IT WAS THE FIRST TIME I'D USED STRATTON BUT I HAD MADE ENQUIRIES PREVIOUSLY
IN THE END WE SOLD PRIVATELY & BOUGHT A NEW CAR
WHEN I CAME HOME... I HADN'T PURCHASED A CAR AT THAT POINT. I THOUGHT STRATTON
I KEPT GETTING AN EMAIL FROM THEM [STRATTON] ONCE A MONTH
STRATTON DID A BETTER FINANCE DEAL THAN WHAT KIA DID
STRATTON WAS PROMPT. IT WAS THE DEALER THAT WAS SLOW
STRATTON NOT WELL KNOWN OUTSIDE OF CAR FINANCE
DIDN'T BUY IN THE END; THE DEAL WITH KIA WASN'T SUFFICIENT: LOW TRADE IN VALUE
HAD STRATTON BEEN ABLE TO GET ME A TRADE IN PRICE FOR MY CAR. WD HV BEEN
100% FINANCE IS IMPORTANT BECAUSE I USE MY CASH TO DO SOMETHING ELSE
WHEN I DECIDED TO BUY A CAR THE PEOPLE AT KIA GAVE ME A QUOTE
RE: END CAR I WANTED TO LOWER MY REPAYMENTS & I WANTED A BIGGER CAR
I LOOKED AT CARSALES FOR CAR BECAUSE THEY HAVE THE BEST SELECTION
TIME IS MY MOST VALUABLE RESOURCE
EVEN THOUGH THE DEALERSHIP OFFERED ME A BETTER INTEREST RATE DIDN'T WANT TO DEAL WITH DLR NOW
DEALERSHIP WAS TRYING TO WIN MY BUSINESS WITH FINANCE
I DIDN'T HAVE MULTIPLE CHOICES. CONSULTANT RECOMMENDED ONE OPTION
IT DIDN'T FEEL LIKE AN APPLICATION PROCESS RE: WITH CONSULTANT
SHE TEST DRIVED A COUPLE OF CARS B4 FORE SHE CHOSE ONE
JUST DON'T MAKE ME JUMP THROUGH ALL THE APPROVAL HOOPS
YOU KNOW I'VE GOT GOOD CREDIT, YOU KNOW I PAY ON TIME
I ONLY CONSIDERED BUYING FROM A DEALER WHO LET ME DO A TEST DRIVE
I KNOW WHAT I WANT TO BUY. THIS IS WHAT I WANT TO KEEP MY PAYMENTS AT
CONSULTANT DID NOT RELAY BACK IMPORTANT INFO. TOO FOCUSSED ON BEING FAST
RE: WHAT SHARING CONSULTANT INFO I ENDED UP IN A CONTRACT THAT DIDN'T REFLECT MY GOALS
CAME BACK TO STRATTON BECAUSE ALL DETAILS/INFO P ENTERED SVD
NO CREDIT HISTORY AS SHE IS ORIGINALLY FROM THE [redacted] STATES
HAS HAD MULTIPLE DEALINGS WITH STRATTON
WORKS AS A UX CONSULTANT/ CONTENT STRATEGIST
RELIES ON CONSULTANTS, DID NOT CARE ABOUT TYPE OF FINANCE SHE WOULD HVE
I CANT TELL YOU I WOULD USE STRATTON AGAIN
HAD TO DEAL WITH MULTIPLE CONSULTANTS. THE FIRST GUY LEFT
PREFERS TALKING OVER THE PHONE THAN COMPLETING APP ONLINE
UPSET; MONEY WITHDRAWN FROM HER CARD EARLIER THAN WAS NEGOTIATED
FIRST POSITIVE & SECOND NEGATIVE
DISLIKES PUSHY SALES CONSULTANTS
SKEPTICAL OF CERTAIN Q. BEING ASKED. [redacted]


RAY
KELVIN
SAM
BIO
OWNS 12 HOUSES & EARNS BY COLLECTING MORTGAGE
SITES/BRANDS - CAR SALES - PICKLES AUCTIONS
STRATTON NOT WELL KNOWN OUTSIDE OF CAR FINANCE
SOLE PROPRIETER - REAL ESTATE
WORKS AS A UX CONSULTANT / CONTENT STRATEGIST
NO CREDIT HISTORY AS I'M ORIGINALLY FROM THE STATES
GOALS / MOTIVATION
EVERY TIME I SHOP AROUND FOR FINANCE, THAT'S HOW MOST OF MY TIME IS SPENT
I'M LOOKING FOR A GOOD TRADE IN DEAL FOR MY OLD CAR
I CHOSE KIA BECAUSE OF THE 7 YR WARRANTY. I KEEP FOR 5 YRS THEN IT'S OVER.
LOOKING FOR DETAILED QUOTE, INTEREST RATES, RESIDUAL REPAYMENT PERIODS
STRATTON WAS PROMPT. IT WAS THE DEALER THAT WAS SLOW
I DON'T WANT TO TALK TO ANYONE WHEN GETTING A QUOTE
I LIKE TO COMPARE DIFFERENT QUOTES BEFORE CHOOSING
I LIKE TO SEE LOAN REPAYMENTS DISPLAYED MONTHLY
DOESN'T MATTER WHO I DEAL WITH, AS LONG AS THEY COME UP WITH THE BEST DEAL!
HAD STRATTON BEEN ABLE TO GET ME A TRADE IN PRICE FOR MY CAR, SO WE'D BE
EXIT & INT... MAIN CONCERN
'I CHANGE MY CAR EVERY 3 YEARS'
'HE [CONSULTANT] HELPED ME ALOT... OK, I HAVE TO GIVE BACK' [W/BIZ]
I DIDN'T BOTHER SHOPPING AROUND. STICK WITH STRATTON
MORE EXPENSIVE, BUT BECAUSE OF THE SERVICE
GIVE ME A QUOTE AS FAST AS POSSIBLE
I DON'T WANT TO BE LOCKED DOWN WITH EXIT FEES
TOUCH POINTS INTERACTION
I WAS LOOKING FOR A CAR 2-3 YEARS AGO
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I KEPT GETTING AN EMAIL FROM THEM [STRATTON] ONCE A MONTH
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WHEN I CAME HOME... I HADN'T PURCHASED A CAR AT THAT POINT. I THOUGHT STRATTON
PREFERS TALKING OVER THE PHONE THAN COMPLETING APP ONLINE
RELIES ON CONSULTANTS. DID NOT CARE ABOUT TYPE OF FINANCE SHE WOULD HAVE
RE: END CAR. I WANTED TO LOWER MY REPAYMENTS & I WANTED A BIGGER CAR
CAME BACK TO STRATTON BECAUSE ALL DETAILS/INFO PREVIOUSLY ENTERED SAVED
TIME IS MY MOST VALUABLE RESOURCE
I KNOW WHAT I WANT TO BUY. THIS IS WHAT I WANT TO KEEP MY PAYMENTS AT
I JUMPED ON WEBSITE/ REQUESTED QUOTE
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STRATTON DID A BETTER FINANCE DEAL THAN WHAT KIA DID
I THINK I GAVE THEM THE DEAL WHEN KIA AND SAID BEAT IT!
IN THE END WE SOLD PRIVATELY & BOUGHT A NEW CAR.
I ONLY CONSIDERED BUYING FROM A DEALER WHO LET ME DO A TEST DRIVE
IT WAS THE FIRST TIME I'D USED STRATTON BUT I HAD MADE ENQUIRIES PREVIOUSLY
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PAIN POINTS
HAD A HARD TIME READING SMALL TEXT
DIDN'T BUY IN THE END; THE DEAL WITH KIA WASN'T SUFFICIENT. LOW TRADE IN VALUE
AWARE FINANCIAL CANDESCAPE IS ALWAYS CHANGING
I'M SELF EMPLOYED. IT'S OFTEN COMPLICATED IN RELATION TO FINANCE
RE: FINANCE... APR, HOW DO YOU ACTUALLY CALL THEM PRICES?
DEALERS OBNOXIOUS. THEY WANT ME TO COMMIT TO CAR BEFORE SPK TO ME
'LETS RE-CHECK CONFIRM FIRST, THEN WE TALK FINANCE'
HAD TO DEAL WITH MULTIPLE CONSULTANTS. THE FIRST GUY LEFT
CONSULTANT DID NOT RELAY BACK IMPORTANT INFO. TOO FOCUSSED ON BEING FAST
UPSET; MONEY WITHDRAWN FROM HER CARD EARLIER THAN WAS NEGOTIATED
I CAN'T TELL YOU I WOULD USE STRATTON AGAIN
DISLIKED PUSHY SALES CONSULTANTS
EVENTHOUGH THE DEALERSHIP OFFERED ME A BETTER INTEREST RATE DIDN'T WANT TO DEAL WITH ONLINE
JUST DON'T MAKE ME JUMP THROUGH ALL THE APPROVAL HOOPS
RE: NOT SIGNING ANYTHING UNTIL I ENDED UP IN A CONTRACT THAT DIDN'T REFLECT MY GOALS
SKEPTICAL OF CERTAIN Q. BEING ASKED.
STRATTON LOST DEAL TO CAR COMPANY BECAUSE THEY OFFD BTR RTS
FOUND STRIN ONLINE THROUGH GOOGLE SEARCH
REPAYMENT SCHED/OPTIONS WHICH FINANCE COMPANY
FOUND STRATTON ONLINE DID AN ONLINE QUOTE
IT WAS VERY EASY. SOMEONE LEAD ME THROUGH THE WHOLE PROCESS
CC - TENDER PROCESS; PRETTY GOOD DEAL!
GOT 2 OTHER CARS AFTER FIRST, THRU STRATTON
KEEN TO GET SECOND CAR THROUGH STRIN THOUGH
I DIDN'T HAVE MULTIPLE CHOICES. CONSULTANT RECOMMENDED ONE OPTION
HAS HAD MULTIPLE DEALINGS WITH STRATTON
IT DIDN'T FEEL LIKE AN APPLICATION PROCESS WITH CONSULTANT
DEALERSHIP WAS TRYING TO WIN MY BUSINESS WITH FINANCE
TEST DRIVED A COUPLE OF CARS BEFORE CHOOSING ONE
I LOOKED AT CARSALES FOR CAR BECAUSE THEY HAVE THE BEST SELECTION
FIRST POSITIVE & SECOND NEGATIVE
YOU KNOW I'VE GOT GOOD CREDIT. YOU KNOW I PAY ON TIME

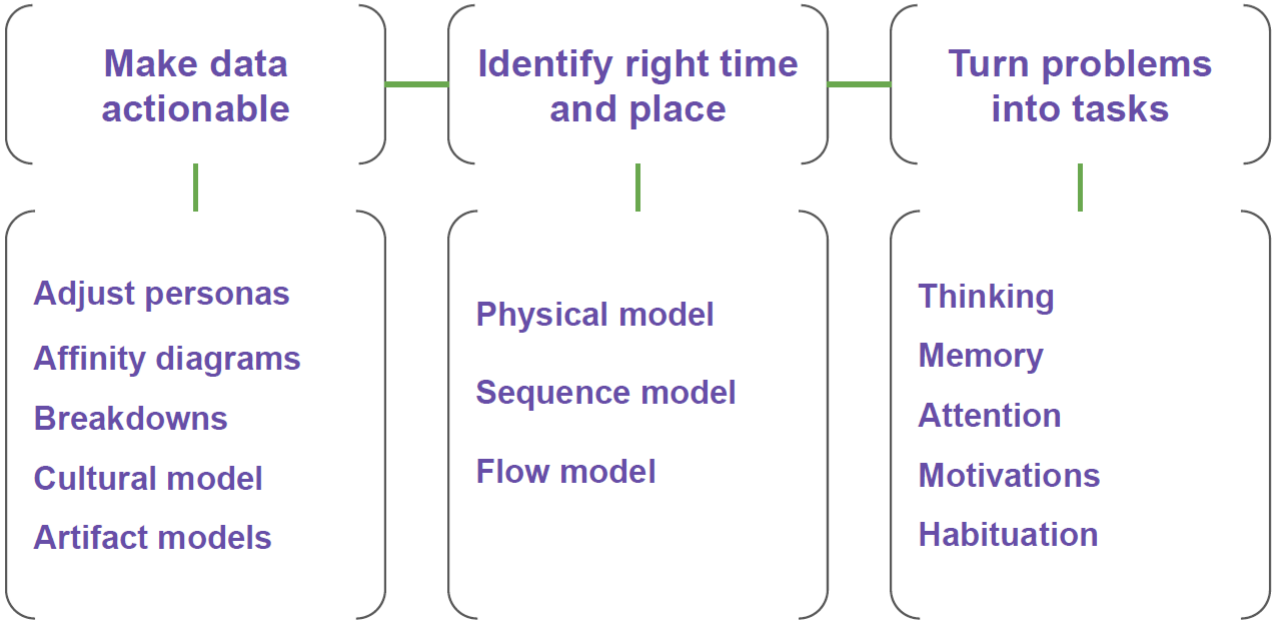


Work models


Understand your users
Translate needs into functionalities
Create design ideas
Prototype design
Test & Evaluate
Build final version



Translating Needs Into Functionalities



Work Model

- A graphical way of representing the data
- Representing the key aspects of work that design teams need to account for in their designs
- A concrete set of deliverables allowing the design team to agree upon a shared understanding and representation

Types of models

- Flow model
- Sequence model
- Artifact model
- Cultural model
- Physical model

Flow & Sequence

- Sequence: Work tasks are often an ordered sequence of actions, unfolding over time
- Flow: How work is broken up across people, how they coordinate

- Components
  - Time and Place: Where does this occur and over what time scale?
  - Individuals & Groups: bubbles with job titles/group name, annotated with responsibilities (list)
  - Intents: An objective for the individual/group
  - Flow: communication between individuals, groups, artifacts
  - Breakdowns: potential disruptions on communication



- Starting point: intents (i.e., objective)

Flow Model

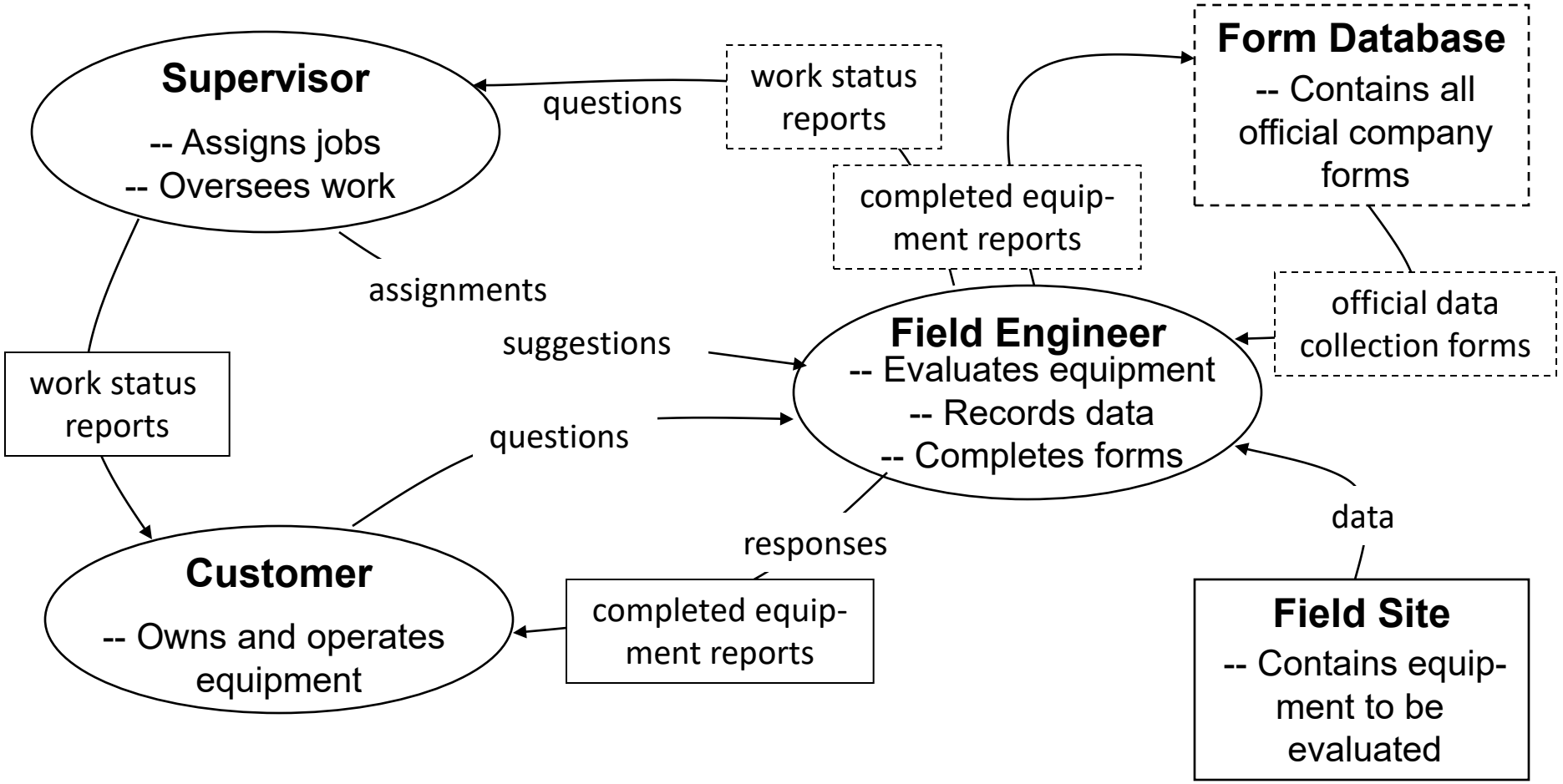
- A bird's-eye view of the organization, showing the people and their responsibilities, the communication path between people independent of time, and the things communicated
- Purpose: Capture and Visualize
  - Roles of individuals
  - Communication and coordination
  - Mechanisms that help and hinder
  - Strategies

- Focuses on the roles of different users, and how they communicate and coordinate to get work done
- Includes the places where communication happens, the artifacts used for communication, and breakdowns in communication that negatively impact work

Components of Flow Model

- Individuals (Bubbles with responsibilities)
- Groups of individuals with common actions
- Places central to work (Large box w/ responsibilities)
- Responsibilities
- Flow (Arrows between individuals)
- Artifacts (Small boxed label on a flow)
- Communication topic or action (Label on a flow)
- Breakdowns in communication (Lightning bolt!)

Flow Model: Engineer



Sequence Model

- "Low-level, step-by-step information on how work is actually done"
- Includes the intent behind the action, the trigger that led the user to this action, and breakdowns that create problems
- Captured at level of detail appropriate for focus of the design team

- Examines the actions of workers, and how/when they occur
- By examining their structure, you can identify:
  - Their strategy
  - Their intent
  - Important dynamics of the task
- Related to 'procedures'

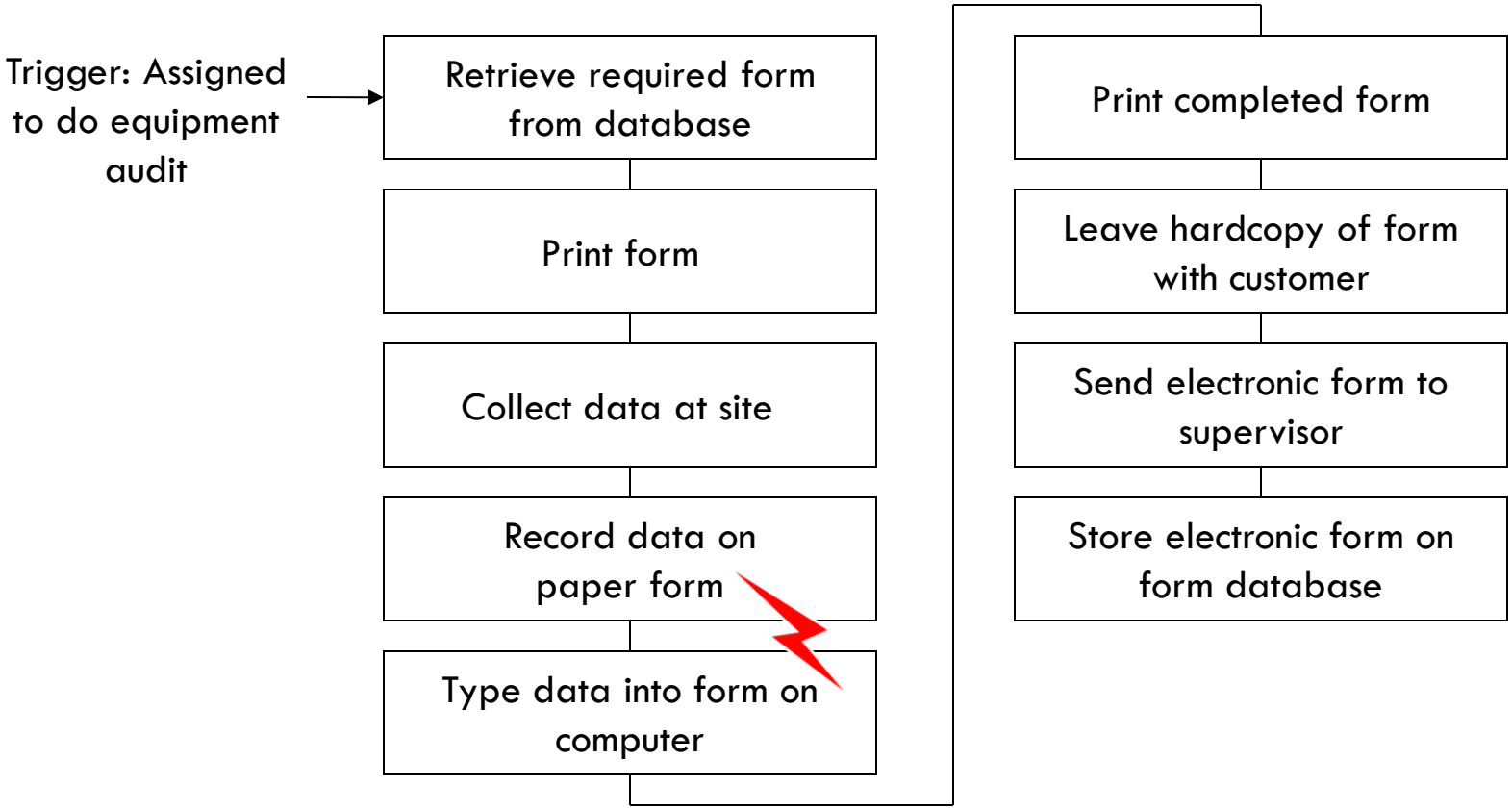
Sequence Model

- Useful for laying out sequence of:
  - Actions
  - Processes
- Similar to
  - Process charts
  - Work cards
- But is descriptive, not normative!

Components of Sequence Model

- Intent
- Trigger
- Steps
- Order
- Breakdowns

Sequence Model: Database



Cultural Model

- "Work takes place in a culture, which defines expectations, desires, policies, values, and the whole approach people take to work"
- Revealed in the language used to describe work, the tone of the place, the policies, and the influence of the overall organization
- Influencers are the individuals, formal groups or abstract principles that influence the work of specific people

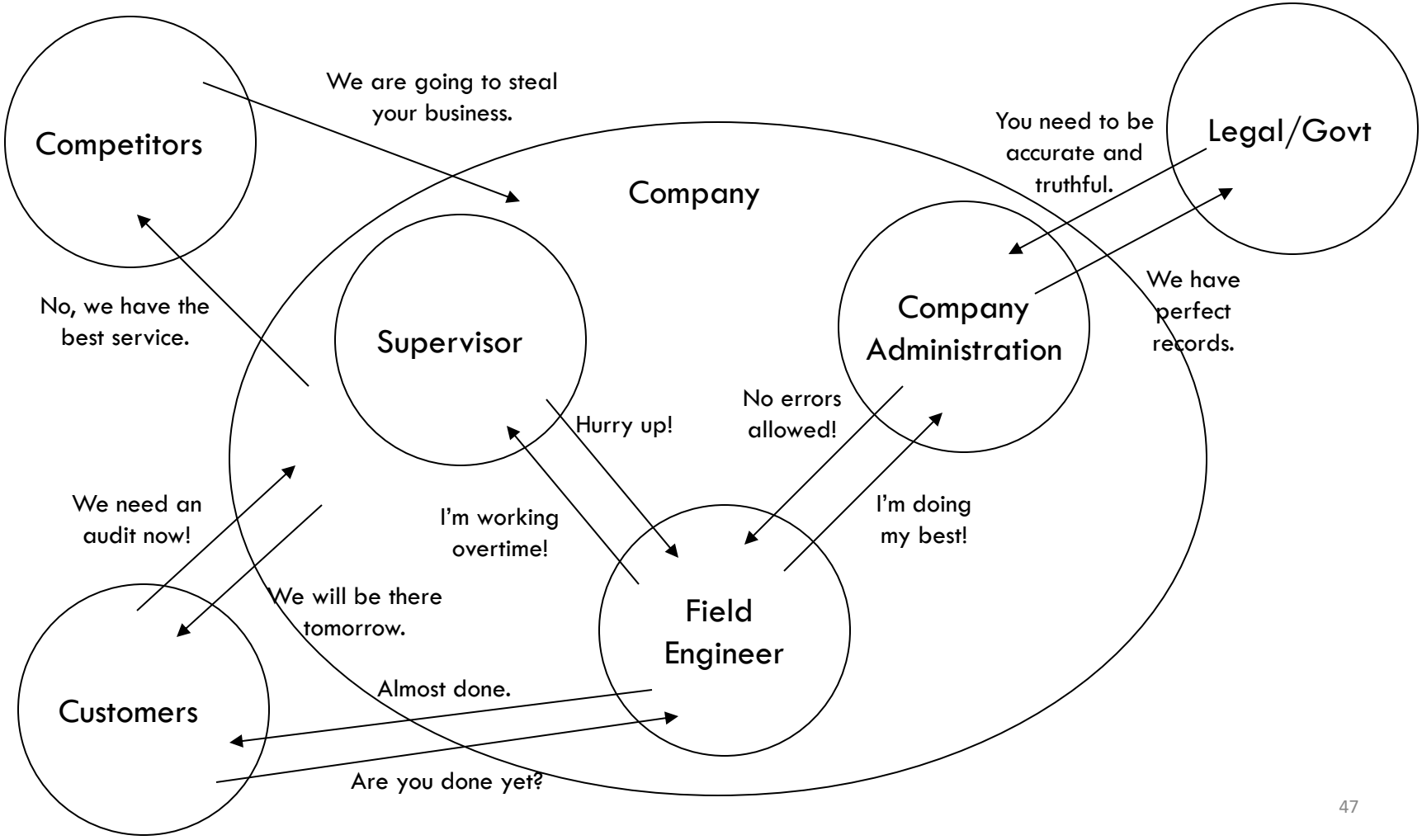
Importance of Cultural Model

- Your end-product might work well in theory, but in practice wouldn't be used because of
  - Differing expectations
  - Specific desires
  - Policies
  - Values
  - Changes in approach at organizational level

Components of Cultural Model

- Influencer
- Extent
- Influence
- Breakdown

Cultural Model: Developer



Artifact Model

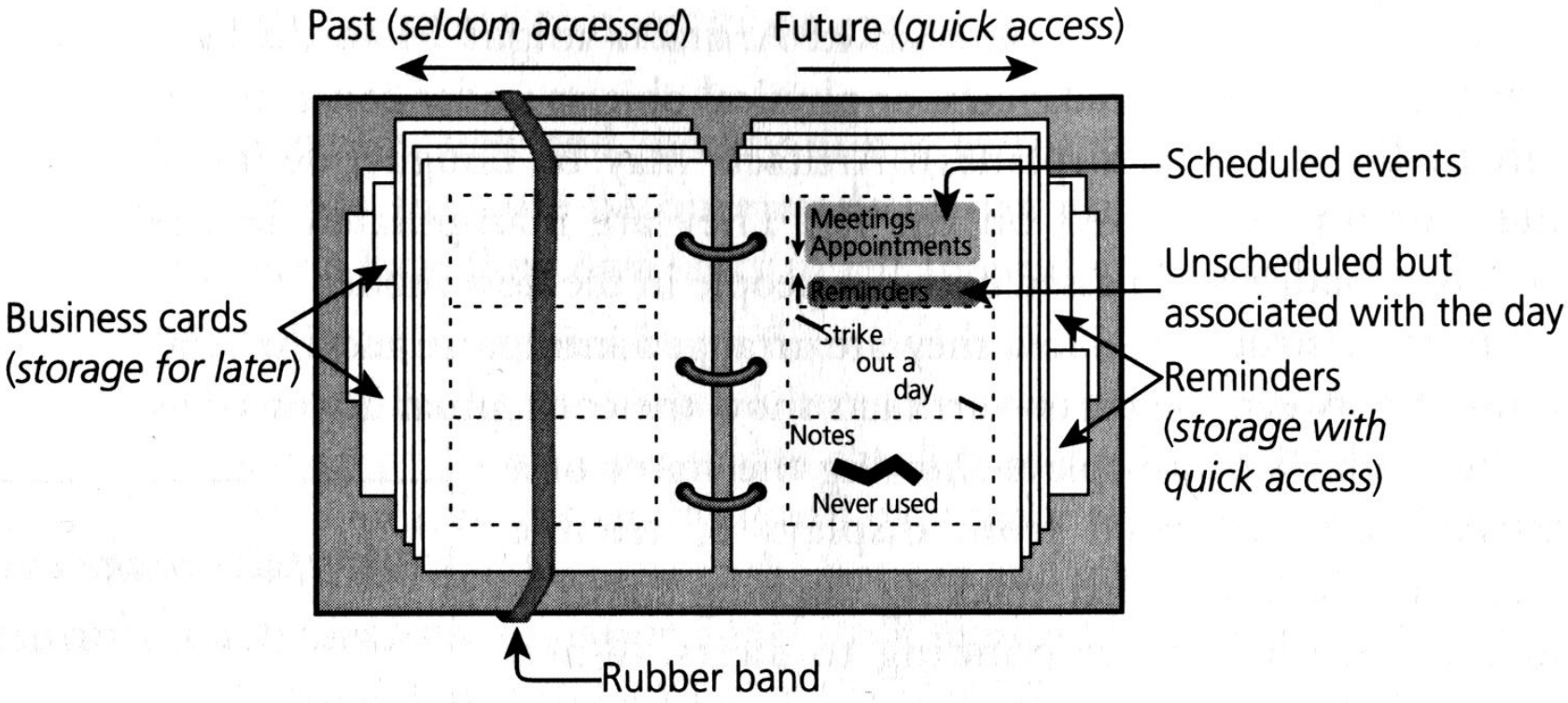
- What people create, use, modify things during work
  - Tool or end-object
  - 'Official' or 'Informal'
- Way of working can be built right into artifact
  - Reveals strategies, required structures, information

- "An artifact model is a drawing or photocopy of the artifact, complete with any handwritten notes"
- Documents physical artifacts used in work
- During the CI, interviewers should inquire into the structure, content, presentation and usage of the artifact, as well as any breakdowns in its current use

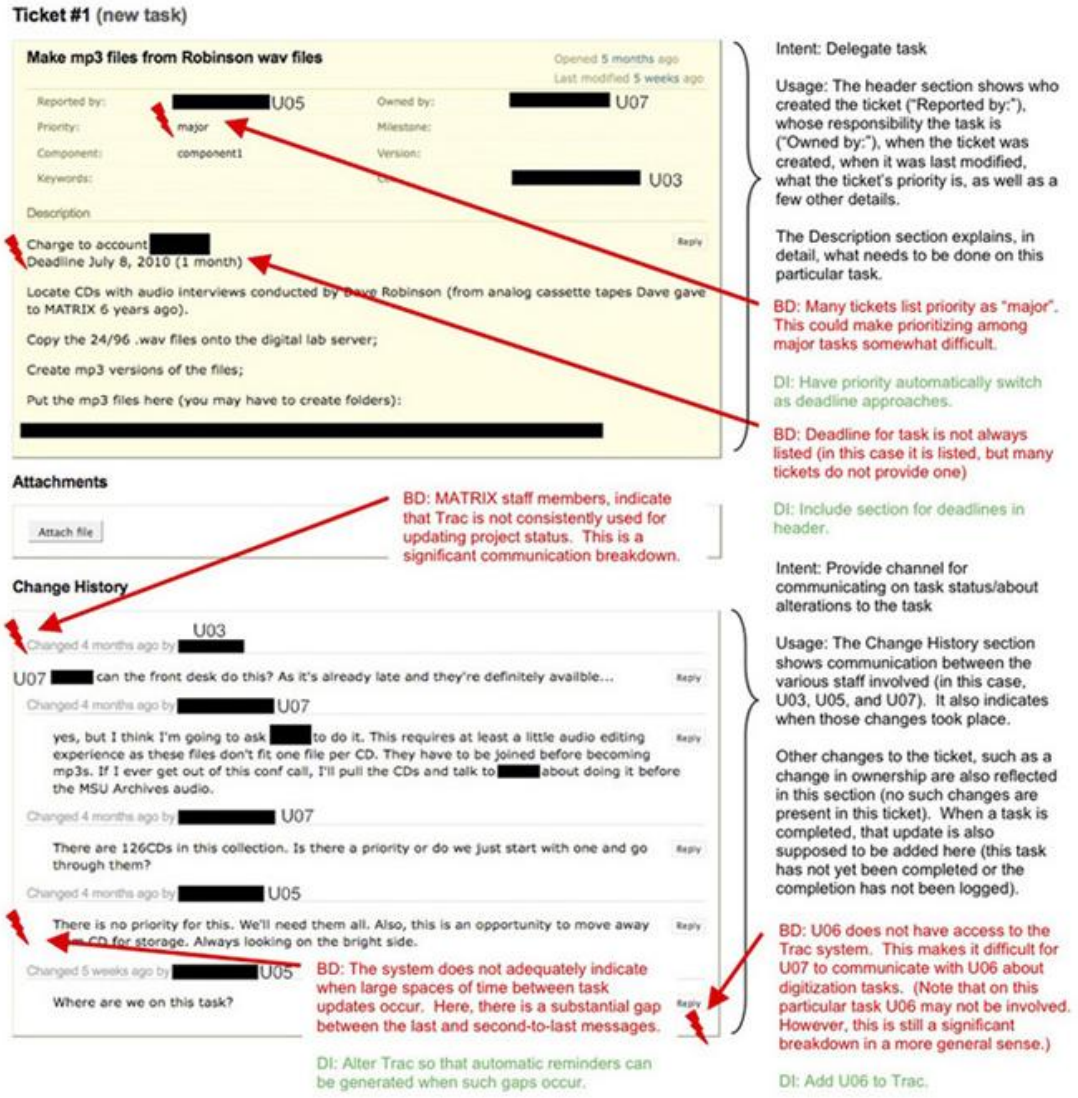
Components of Artifact Model

- Information
- Parts of the object
- Structure of the parts
- Annotations
- Presentation of the object
- Usage of the artifacts
- Breakdowns

Artifact Model: Personal Calendar



Artifact Model: Ticket tracker



Physical Model

- Documents the physical environment where work happens
- Includes the organization of space, the grouping of people, and their movement in the space
- Focuses on aspects relevant to the work / focus, and not on complete fidelity

Components of Physical Model

- Places where work occurs
- Physical structures that limit and define space
- Usage and movement
- Infrastructure and communication lines
- Artifacts
- Layout
- Breakdowns

Note: Physical model is a caricature of workplace, not a floor plan!

Physical Model: Office

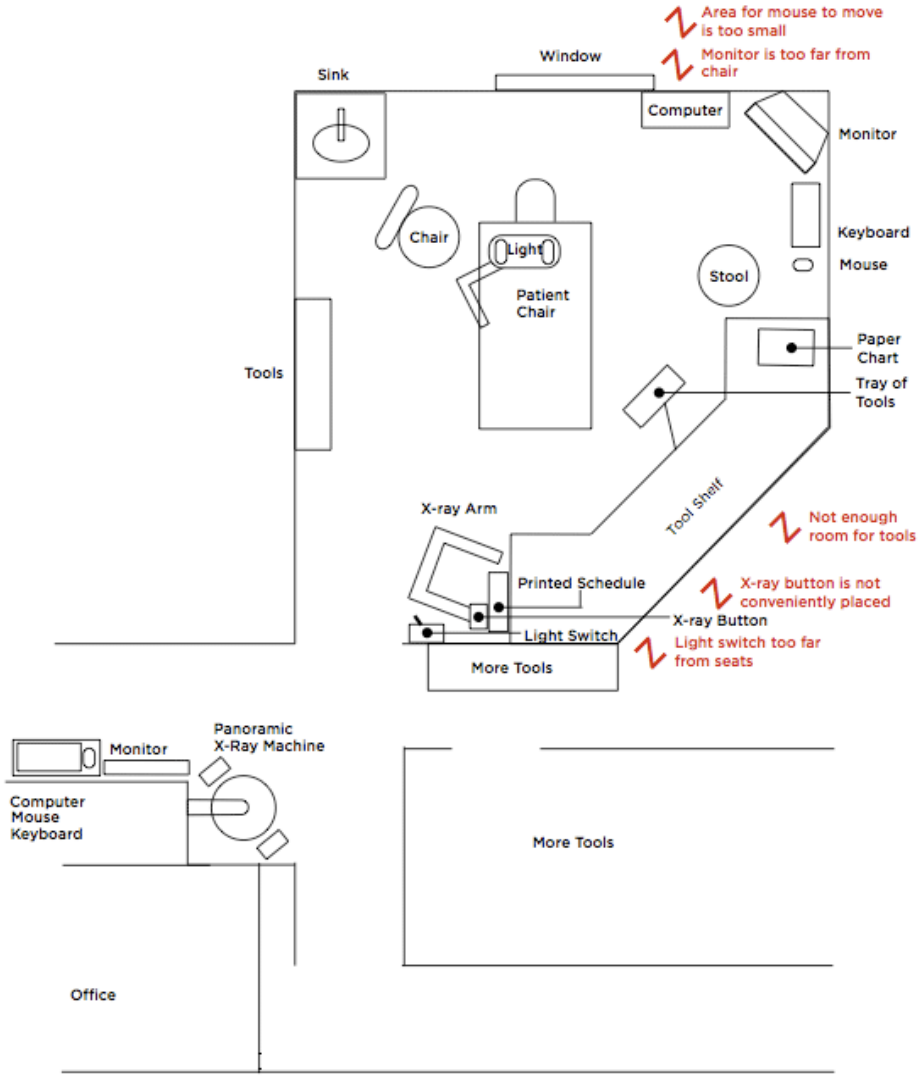
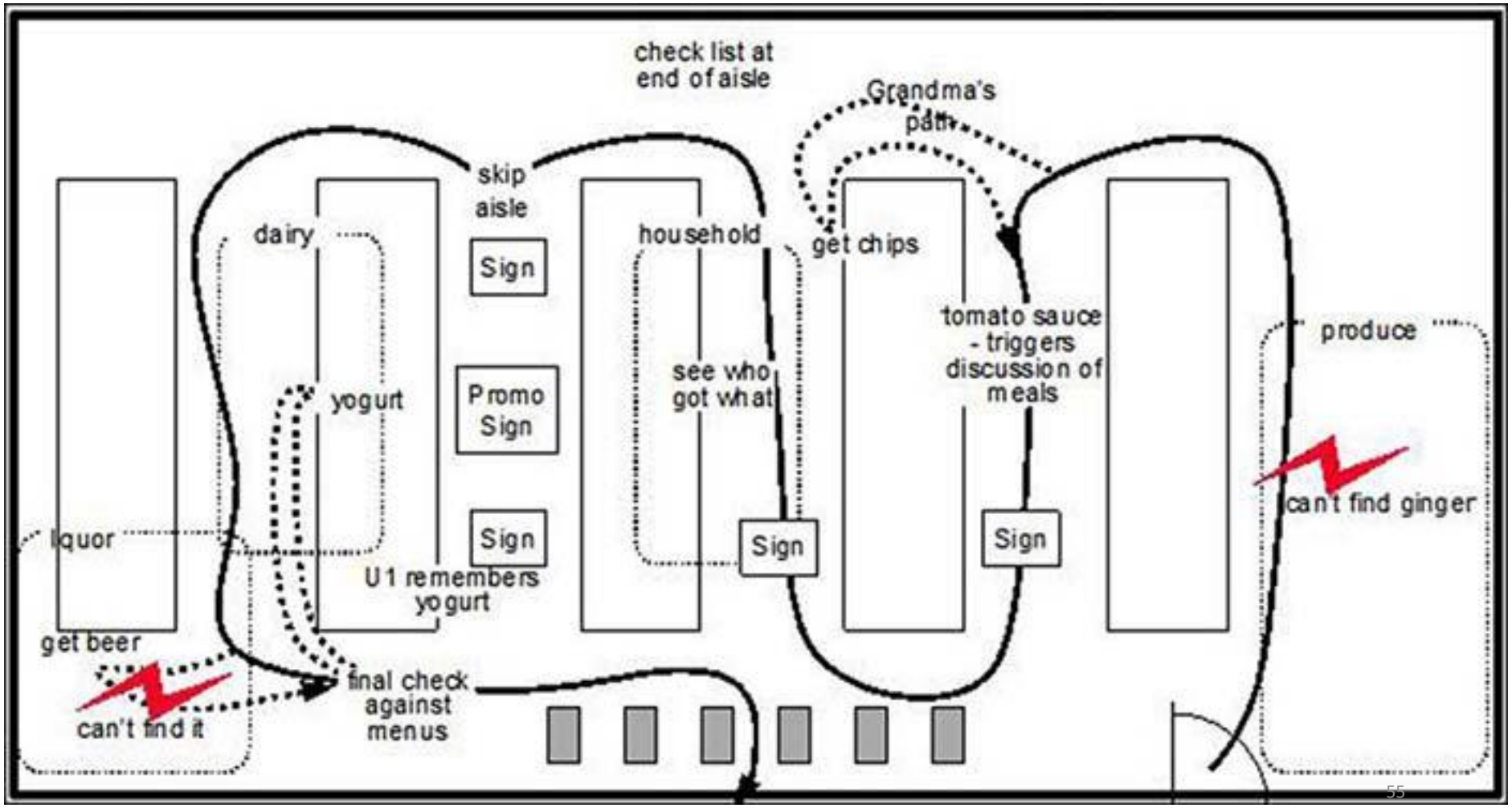
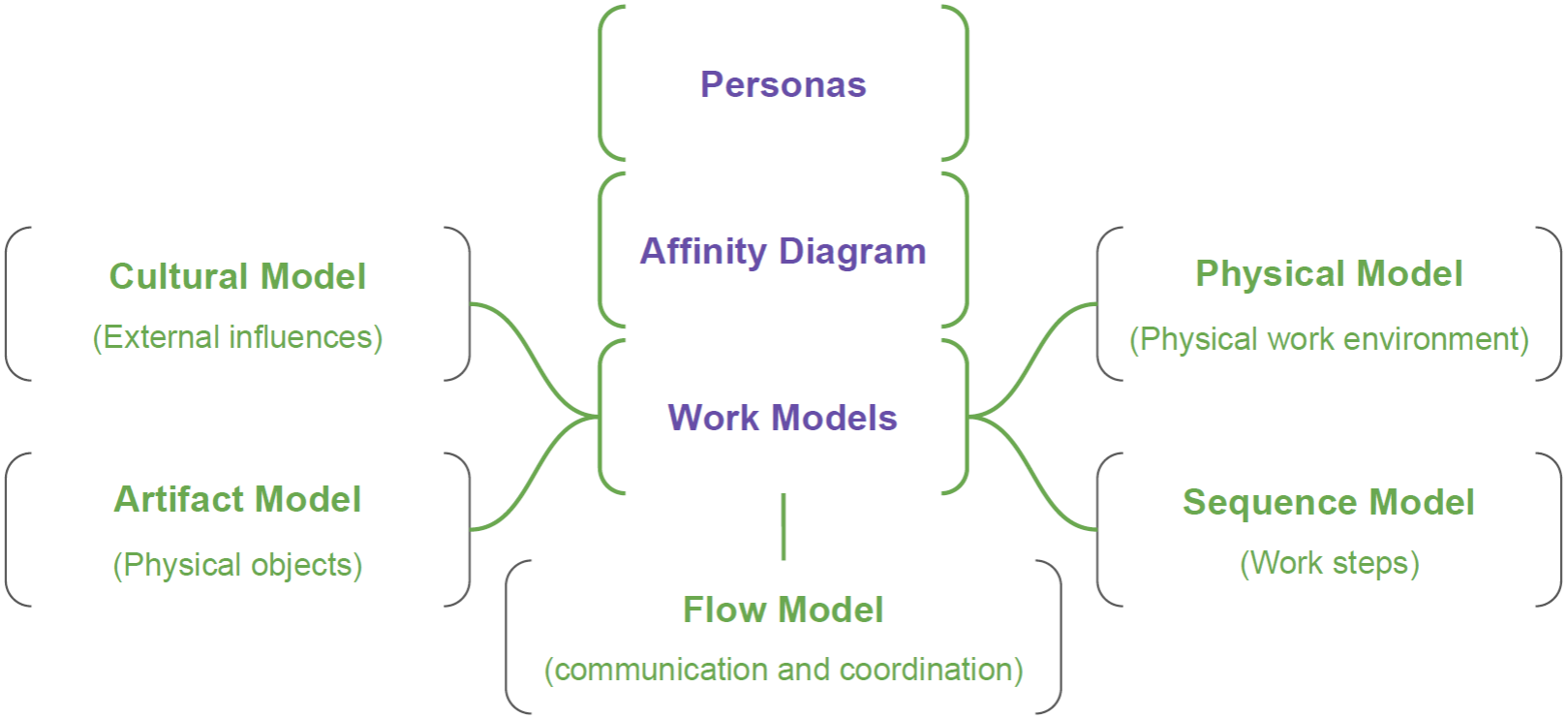


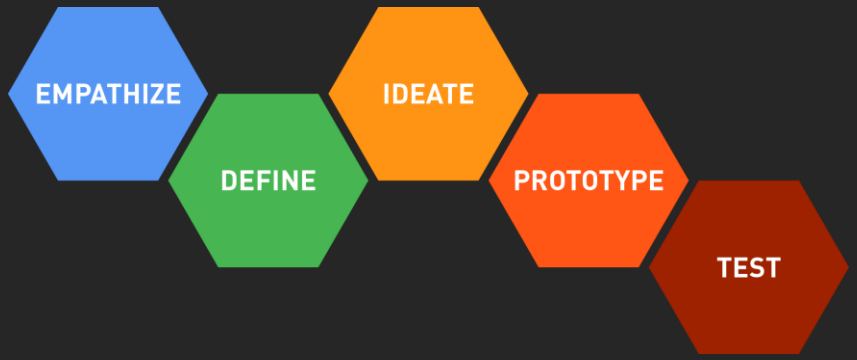
Figure 8: The consolidated physical model

Physical Model: Cafeteria



Translating Needs Into Functionalities





Survey Report

1. Analysis of Quantitative (Survey) Data

Do you want to report…

- how many people answered a, b, c, d?
- the average number or score?
- a change in score between two points in time?
- how people compared?
- how many people reached a certain level?

Common descriptive statistics

- Count (freq)
- Percentage
- Mean
- Mode
- Median
- Range
- Standard dev
- Variance
- Ranking

a. How many people answered a, b, c, d?

b. The average # or score?

c. A change in score between two points in time?

d. How people compared?

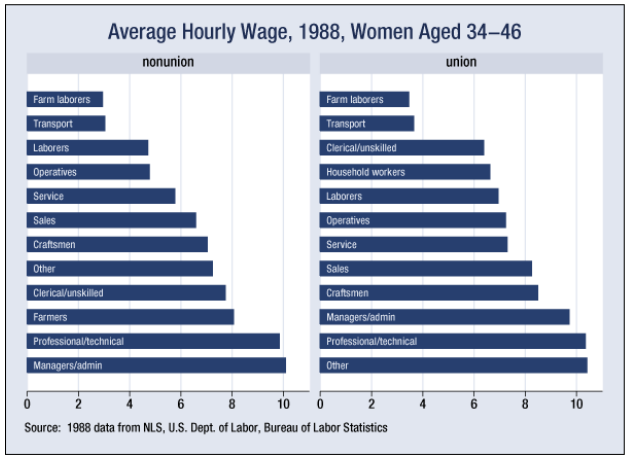
e. How many people reached a certain level?

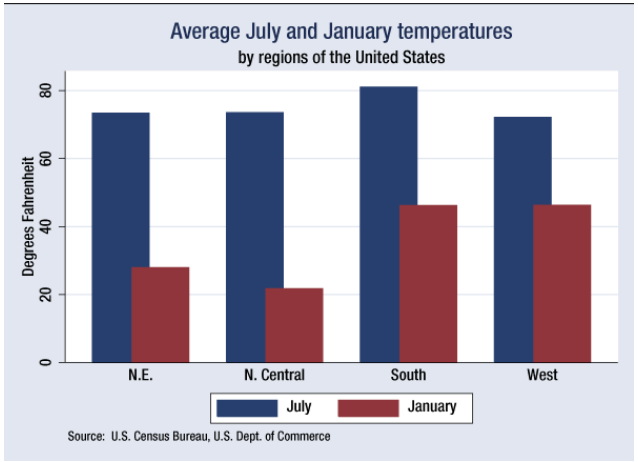
2. Interpret & Visualize data

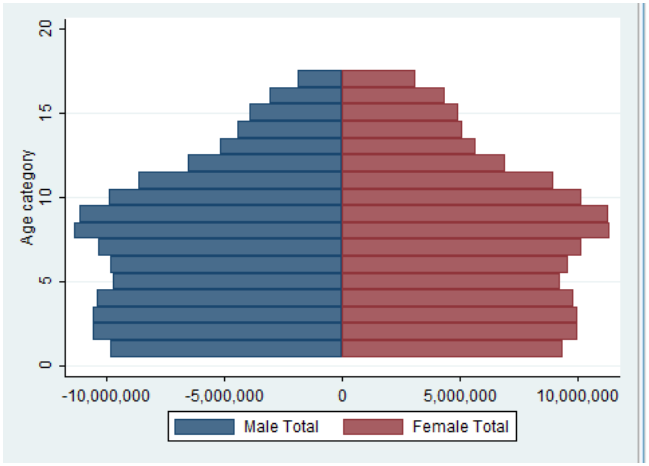
Dig bigger!

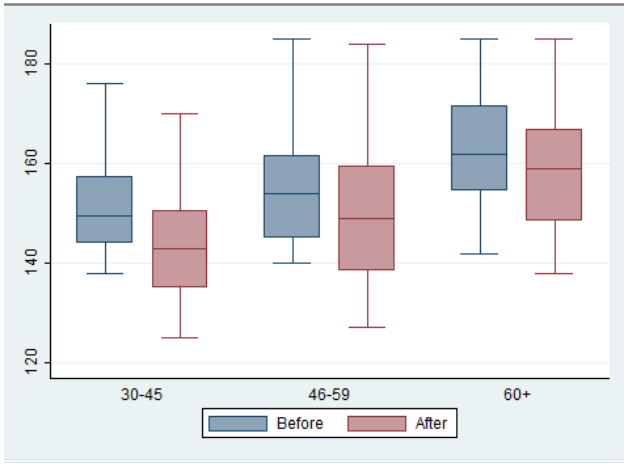
- Did different groups show different results?
- Were there findings that surprised you?
- Are there things you don't understand very well – further study needed?

Visual representations (Graphs)









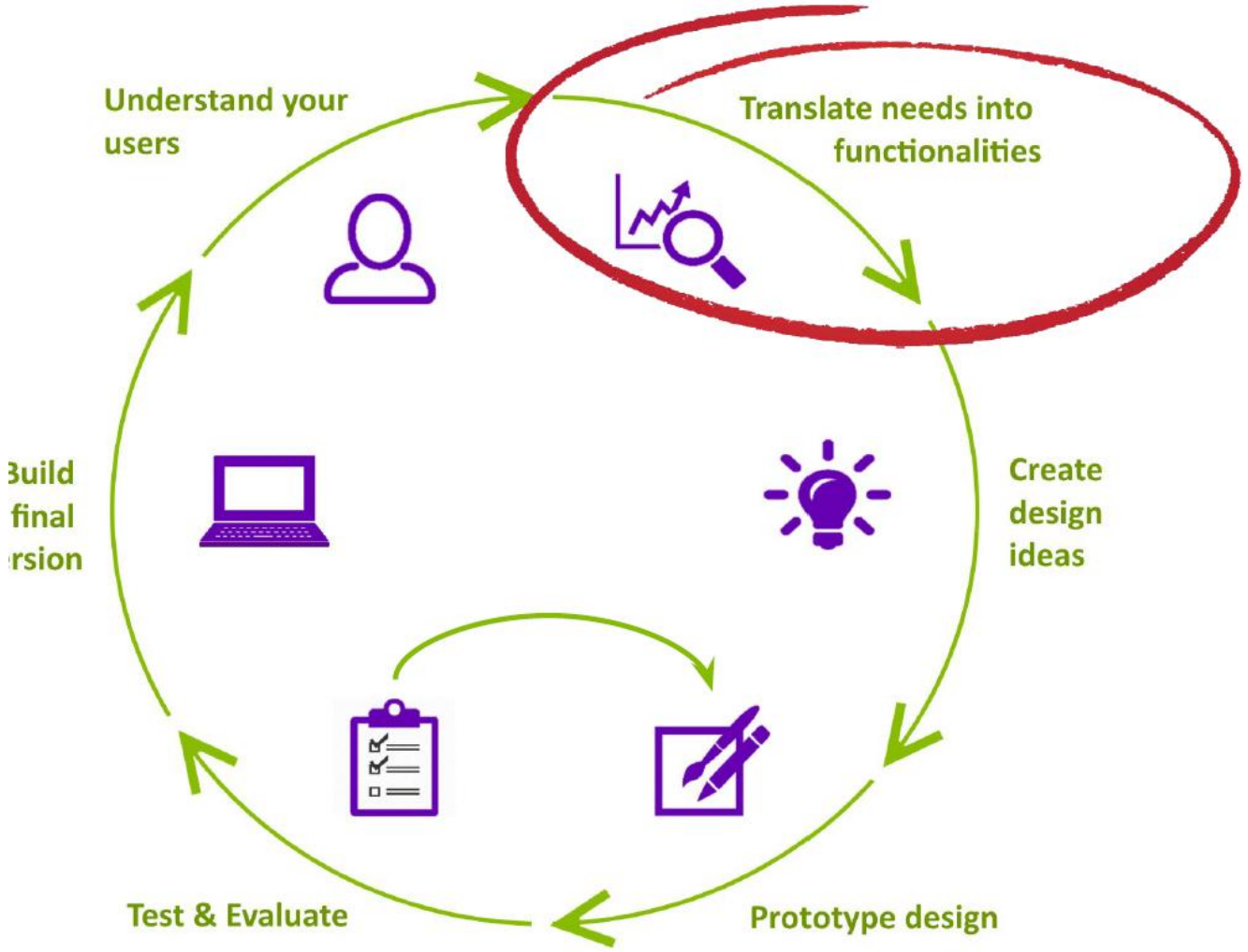
Report: Interview & Survey

Reflect

- What did we learn?
- What conclusions can we draw?
- What are our recommendations?
- What are the limitations of our analysis?

Report

- Be explicit about your limitations
- Know the claims you cannot make
- Do not claim causation without a true experimental design
- Do not generalize to the population without random sample and quality administration (e.g., <60% response rate on a survey)


Understand your users
Translate needs into functionalities
Create design ideas
Prototype design
Test & Evaluate
Build final rsion

Group project

P4. Data Analysis

In this assignment, you will analyze the collected data.

1. Survey data analysis
    1) Report important findings using relevant visualization formats
    2) Report demographic data using descriptive statistics
    3) Write a paragraph to describe what you learned: themes, categories, things that are unexpected or surprised you
2. Interview data analysis: Crate an affinity diagram (30+ sticky notes) --> Turn into a digital version.
    1) Record your diagramming process (photos): Report the final output (synthesis) as well as the process (raw data)
    2) Report a digital version of a final affinity diagram
    3) Write a paragraph describing what you learned: themes, categories, things that are unexpected or surprised you

Due by midnight 10/22 (Saturday)

#Disclaimer. Further instruction of this submission can be given verbally during class or through Piazza.

P4. Data Analysis

Submission format:

- A PDF file, 12 point scale in Times New Roman, 1.5 line spacing
- Add a team name, a description of your project, a link to your project website on top (Your website should be accessible with full contents)
- Body should include:
  1. Survey analysis
  2. Affinity diagram with a process
- At the end: Indicate who contributed to the original writing of each section and who helped improve and/or review at the end

#Disclaimer. Further instruction of this submission can be given verbally during class or through Piazza.

P4. Data Analysis

Rubric

- Survey (2pt)
  - If the survey analysis is presented using a proper visualization format with proper statistical measures, both demographic info and other findings (1pt)
  - If the description clearly explains what you learned (1pt)
- Affinity diagram (5pt)
  - If the final diagram contains significant themes (2pt)
  - If the final diagram has a structure/relation of top and sub themes (1pt)
  - If the submission includes the photos to show the extract-synthesis process and a final digital version (1pt)
  - If the description clearly explains what you learned (1pt)

#Disclaimer. Further instruction of this submission can be given verbally during class or through Piazza.

P4. Data Analysis

Rubric

- You will lose 10% if your website doesn't work
- You will lose 50% if your submission does not follow the format
- You will lose 30% if it's a late submission (Not accepting a late submission submission after 24 hours from the due)

#Disclaimer. Further instruction of this submission can be given verbally during class or through Piazza.